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Credit Risk Drivers in Hotel Sector

The importance of hotels' location and distances to touristic
points

Luis Francisco Matias Silvano

Master Thesis

presented as partial requirement for obtaining a Master's Degree in Statistics and Information Management

NOVA Information Management School
Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

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The importance of hotels' location and distances to touristic points

by

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Master Thesis presented as partial requirement for obtaining the Master's Degree in
Statistics and Information Management, with a specialization in Risk Analysis and
Management.

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism or any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

Luis Silvano

Lisbon, July (14th), 2025

ABSTRACT

This thesis examines the determinants of credit risk in the Portuguese hotel sector, an important segment of the national tourism industry. Using data from multiple sources, including Turismo de Portugal and Orbis, it analyzes individual hotel entities between 2010 and 2015. Credit risk in this research is using a Z-score based on financial indicators, and its relationship with both distances to touristic places and financial variables through panel data regression models. The research highlights that location plays a significant role in reducing credit risk, with hotels situated closer to major tourist points such as beaches, castles, and churches generally exhibiting lower risk profiles. The analysis also extends to traditional financial metrics, showing that hotels with higher efficiency ratios, higher occupancy levels, and a greater number of rooms for events are associated with decreased credit risk. These findings provide useful insights for financial institutions, investors, and policymakers seeking to understand risk exposure in the hotel industry.

KEYWORDS

Credit Risk; Z-score; Locations; Hotel Sector; NUTS

Sustainable Development Goals (SDG):

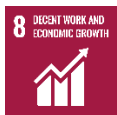


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LIST OF ABBREVIATIONS AND ACRONYMS

API	Application Programming Interface.
AR	AutoRegressive model.
BMEs	Big, Macro Enterprises.
BvD	Bureau van Dijk, provider of the Orbis company financial database.
EU	European Union.
GDP	Gross Domestic Product.
GIS	Geographic Information System.
GMM	Generalized Method of Moments.
IMF	International Monetary Fund.
INE	Instituto Nacional de Estatística, Portugal's national statistical authority.
MSMEs	Micro, Small, and Medium Enterprises.
NACE	Nomenclature of Economic Activities.
NUTS	Nomenclature of Territorial Units for Statistical Purposes.
RNET	Registo Nacional de Empresas de Turismo, the National Registry of Tourism Enterprises.
ROAA	Return on Assets Average.
TINs	Tax Identification Numbers.
UNWTO	United Nations World Tourism Organization.
VAT	Value Added Tax.

1. INTRODUCTION

The hotel sector is a fundamental pillar of the global travel and tourism industry, significantly contributing to economic growth, employment, and international trade. In 2023, destinations around the world earned 1.5 trillion dollars in international tourism receipts (World Tourism Organization, 2024). Tourism directly accounted for 3% of global GDP, amounting to 3.4 trillion dollars.

In the European context, the region remains the world's largest tourism destination. Europe welcomed approximately 707 million international arrivals in 2023 (World Tourism Organization, 2024). This notable performance underscores the hotel sector's critical importance to the region's economic and social development.

Focusing on Portugal's travel and tourism sector has been a significant contributor to the country's gross domestic product (GDP), with the hospitality industry playing a central role in this economic domain. Travel and tourism represented 20% of Portugal's GDP in 2023, particularly through the production of final goods and services (World Travel & Tourism Council 2024 Economic Impact Research). To sustain growth and development, enterprises in this sector require substantial investments. Consequently, credit risk directly affects a variety of stakeholders, including investors, financial institutions, and policymakers. Banks must carefully evaluate historical data to mitigate the risks of non-performing loans, which arise when stakeholders fail to meet repayment obligations. In the context of the hotel sector, this involves considering a combination of factors, including its sensitivity to economic cycles and fluctuating demand patterns, substantial investments in fixed assets, and how low occupancy rates may compromise service quality. Moreover, hotels are particularly vulnerable to external shocks—such as the COVID-19 pandemic—which raises uncertainty and volatility in client behavior. Other critical dimensions, such as seasonality, location, and market competition, further influence the financial stability and credit risk profile of hotel firms. In particular, the geographic location of a hotel is a significant driver of credit risk, as it directly affects its exposure to demand, accessibility, and proximity to tourist, business, or cultural structures. Hotels located in central areas—such as city centers, coastal regions, or near major tourist attractions like castles, churches, or beaches—tend to benefit from higher and more stable occupancy rates. (Alvarez Leon et al., 2021) support this by showing that proximity to urban and coastal attractions improves hotel performance. Therefore, locational characteristics should be incorporated into credit risk assessments, as they directly influence a hotel's ability to generate stable revenues and maintain long-term financial solvency. By balancing the financial needs of the hotel sector with effective risk management strategies, potential threats can be proactively identified and assessed, thereby enhancing decision-making and ensuring sustainable outcomes.

The objective of this research is to analyze and provide guidance on mitigating risks associated with credit risk by identifying its determinants, thereby contributing to financial stability in the hotel sector, with a focus on the importance of location of the hotels and distances to touristic points. We consider variables related to hotels' location, financial indicators and characteristics. The sample is representative¹ of the entire hotel sector in Portugal. over the period 2010–2015 and contains information on 882 tourist points. The hotel sector and respective business

¹ The hotels' sample consists of 444 micro, small and medium-sized enterprises (MSMEs) and 619 large enterprises (BMEs).

models are well diversified and provide a perfect macroeconomic environment for this type of study given the fact that the country offers a diverse range of tourism types, including cultural tourism, religious tourism, and beach tourism across the entire continental territory and islands.

The period 2010–2015 is significant as it reflects an important macroeconomic cycle of financial distress in the World. The years 2012 and 2013 show a higher credit risk and confirm a downturn in the hotel sector during this period. Among other EU countries, Portugal went through a financial crisis between 2010 and 2014, during which time requested external financial assistance through an IMF-EU bailout package. At the same time, the level of demand for hotel services experienced a significant increase of 9 million arrivals recorded during 2012–2013.

This study contributes to the existing literature by offering detailed insights into how the location of hotels and their proximity to various types of tourist attractions influence credit risk and financial performance. It also provides a comprehensive overview of hotel characteristics, and the diversity of services offered across the sector. By incorporating these spatial dimensions, the research delivers valuable evidence for stakeholders operating in or concerned with the hotel industry. Moreover, the dataset can support future studies that analyze hotel closures in relation to geographic location. In this context, survival analysis emerges as a particularly suitable method, as it captures the timing of closure events and accounts for how different types of locations may influence the likelihood of firm survival—an essential aspect in identifying the determinants of credit risk (Lado-Sestayo et al., 2016). The findings aim to inform decision-making by investors, financial institutions, and policymakers, thereby enhancing strategic planning and promoting more sustainable financial and operational outcomes in the hospitality sector.

The study is organized as follows: Section 2 presents a review of related literature, while Section 3 outlines the data and methodology. Section 4 explains the results and discussion, and Section 5 concludes the study and suggests opportunities for future research.

2. LITERATURE REVIEW

Previous studies on credit risk in the hotel sector have primarily focused on analyzing the characteristics and main determinants of risk (Kim et al., 2012) and the impact of location in the hotel sector (Vivel-Búa et al., 2018). This study expands on the previous analysis by examining different definitions of distances between hotels' locations and touristic points, considering both micro, small, and medium-sized enterprises (MSMEs) as well as large, macro-enterprises (BMEs).²

While previous studies have highlighted the significance of location for hotel performance (Chen & Chang, 2012), (Lado-Sestayo et al., 2016), (Sainaghi, 2011), (Vivel-Búa et al., 2018) they have not thoroughly explored in detail how proximity to tourist points might influence credit risk.

To systematically capture this spatial proximity, this study adopts a geographic information system (GIS)-based approach. Following the methodology applied by (Sholeh et al., 2017), who used latitude and longitude data and the Google Maps API to locate and map business entities, this approach enables our research to calculate precise distances between hotels and key tourist attractions. This process ensures greater accuracy and consistency when incorporating geographic variables into credit risk models.

Moreover, this research considers traditional determinants already described in literature as controls, such as liquidity, leverage, profitability.

One key factor influencing credit risk in the hotel sector is the asset size of the establishments. Larger hotels tend to leverage their market position to gain competitive advantages (Ang et al., 1985) and possess greater capacity to absorb and adapt to economic fluctuations affecting their operations (Sullivan, 1978).

Operational efficiency likewise affects risk levels. Efficient resource utilization enhances the firm's capacity to generate revenues and profits, thereby mitigating credit risk. This effect has been corroborated in studies examining systematic risk within the sector Kim et al. (2002, 2007, 2012).

Profitability is also a relevant factor, as firms with higher returns on assets tend to have a reduced probability of financial distress and bankruptcy (Chen, 2013).

In summary, this study contributes to the literature on credit risk in the hotel sector by: (i) simultaneously examining location and diversification alongside hotel-specific characteristics as risk determinants; (ii) focusing on a sample predominantly composed of micro, small, and medium-sized enterprises (MSMEs) and macro-enterprises (BMEs), analyzing the Portugal hotel market, which is a significant country internationally.

² MSMEs account for 41% of registered hotels in Portugal, while BMEs represent 59%, as reported by the RNET database from Turismo de Portugal.

3. DATA AND METHODOLOGY

We use comprehensive information about the hotels registered in Portugal. Different data sources were used, namely: Turismo de Portugal, Visit Portugal website; Orbis BvD; INE, Statista, and Pordata. The Turismo de Portugal dataset provided extensive details about the characteristics of hotels. Orbis database provided financial information.

We aimed to integrate hotel characteristics with their corresponding financial data. The integration of the datasets was made possible with common variables. For instance, the Tax Identification Number and NUTS II. The financial data from Orbis was matched with the corresponding hotels using their respective Tax Identification Numbers, ensuring accurate and cohesive dataset integration for the analysis. To ensure consistency between the datasets, we calculated the percentage of hotels whose Tax Identification Numbers matched between the Turismo de Portugal dataset and the Orbis dataset, focusing on the Nomenclature of Economic Activities (NACE) codes 55 and classified as legal entities. The analysis revealed a 95.33% compatibility rate, indicating that most hotels in the Turismo de Portugal dataset had corresponding financial data available in the Orbis dataset. This high degree of alignment allowed us to proceed with our analysis without significant loss of data due to unmatched identifiers or missing financial information for the respective hotels. The NUTS II variable facilitated the linkage between the datasets containing information on Portuguese hotels and tourist points.

Sample Representativeness

Data from Turismo de Portugal was used to include all hotels from Portugal that opened in or before 2015, allowing the inclusion of 1,149 hotels (see [Annex 03 – Data preparation](#) for details). The focus of this study was specifically on analyzing hotels registered as firms, i.e. “legal entities” and excluding “individuals”. After filtering, our sample of Portuguese hotels was reduced to 1,063 (i.e. only less 86 hotels).

Then, we categorized those hotels based on their size using the number of rooms (Devesa & Peñalver, 2013) as follows: micro-hotels in case of 20 or fewer rooms; small hotels for those between 21 to 50 rooms; medium hotels for those between 51 and 80 rooms; large hotels for those between 81 to 120 rooms; and macro-hotels for those with more than 121 rooms. Within this sample, filtered by NACE code 55, we identified 14 micro-hotels, 236 small hotels, 194 medium hotels, 180 large hotels, and 439 macro-hotels. Overall, 444 hotels were classified as MSMEs (Micro, Small, and Medium Enterprises), representing 42% of all registered hotels, while 619 hotels were categorized as large and macro enterprises, accounting for 58% of the total. Moreover, we identified 223 hotels classified as part of a chain of hotels and 840 as individual hotels. In this sense, the sample used in this study is representative of the hotel sector allowing to generalize findings to the entire population with a reasonable degree of accuracy.

Distance between each hotel's location and touristic location

We follow (Vivel-Búa et al., 2018), developing the variable **location** (distance, in kilometers, between hotels and tourist attractions, including castles, beaches, and churches). We used the VisitPortugal website to identify all relevant tourist points across the country. This approach ensured that the geographic data collected was both systematic and reliable for further analysis. We developed the following five-distance-based variables. This method aligns with the geographic information system (GIS) approach described by (Sholeh et al., 2017), where the authors demonstrate how the use of **latitude and longitude coordinates** enables the identification, storage, and spatial visualization of tourism-related locations through Google Maps API.

Location 1 (ht_location_1) as the distance between each hotel and the nearest tourist attraction.

Location 2 (ht_location_2) as the average distance to the three nearest tourist attractions.

Location 3 (ht_location_3) as the average distance to the six nearest tourist attractions.

Location 4 (ht_location_4): as the average distance to the nine nearest tourist attractions.

Location 5 (ht_location_5): as the average distance to the twelve nearest tourist attractions.

Distances were calculated using the Euclidean distance (Maria et al., 2020).

$$D = \sqrt{(\phi_2 - \phi_1)^2 + (\lambda_2 - \lambda_1)^2} \quad (1)$$

Where, D = Euclidean distance in degrees, ϕ_i = latitude location i, λ_i = longitude location i., and the distance D in degrees is converted to kilometers by multiplying D by 111,319 km (1-degree earth = 69 miles $\sim$ 111,319 km).

The Euclidean is a heuristic function obtained based on direct distance free of obstacles. Previous research (Vivel-Búa et al., 2018), used also Euclidean method to measure the distance from touristic centers (a specific location that includes and groups several touristic places) to their respective hotels. In our study we use the same distance formula, but with greater precision, by measuring the specific distance from touristic points (and not only touristic center's locations in general) to the hotels.

By strengthening this location methodology, we further enhanced the analysis as mentioned before by classifying tourist attractions into distinct categories: beaches, castles, and churches. In this way, we created again five locations for each category (i.e. ht_location_beach1 to ht_location_beach5; ht_location_castle1 to ht_location_castle5; ht_location_church1 to ht_location_church5, i.e. following the same methodology).

Control Variables

We identified and analyzed detailed information on the characteristics of the Portuguese hotels (source: Turismo de Portugal). [Table 10: Hotel's characteristics](#) presents the available variables regarding characteristics of the hotels and respective definitions. The variables that were used in detail in this study providing the characteristics of the hotels include: star classification (ht_quality_stars), capacity (ht_events_meetings) and supportive variables (specifically room occupancy and bed occupancy rates) to construct additional control variables related to market concentration, seasonality, occupancy, and market share, following the methodology of (Vivel-Búa et al., 2018). We also incorporated regional variables, namely the 2024 NUTS II region and NUTS III region (ht_nuts_2_newcriteria, ht_nuts_3_newcriteria).

Firstly, we created the market share variable by region (mkt_share_nutsii, mkt_share_nutsiii) defined as each hotel's share of total annual revenue within its respective NUTS II and NUTS III region (f_tot_oper_rev_turn_nutsii, f_tot_oper_rev_turn_nutsiii) from 2010 to 2015. Due to confidentiality constraints, hotel-level room availability and occupancy data were not accessible, so regional-level aggregation was required. We aggregated annual revenue data by region. Specifically, total revenues were calculated for each year and geographic level (NUTS II and NUTS III). Secondly, market concentration (mkt_share_conc_nutsii, and mkt_share_conc_nutsiii) was computed by squaring each hotel's market share (mkt_share_pwr_nutsii, and mkt_share_pwr_nutsiii) and summing these squared values by year and region (f_mkt_share_pwr_nutsii, f_mkt_share_pwr_nutsiii). These values were then assigned to each hotel based on its corresponding year and region. Thirdly, the occupancy control variable was developed using monthly room occupancy rates. For each month (January to December), data were collected at region level, i.e. NUTS II (ht_room_occup_rt_nutsii_1 to _12) and NUTS III (ht_room_occup_rt_nutsiii_1 to _12). These monthly values were used to calculate annual averages (ht_room_occup_nutsii and ht_room_occup_nutsiii). These averages were linked to each hotel based on region and year. Finally, to measure seasonality (ht_room_occup_sea_nutsii and ht_room_occup_sea_nutsiii) we used the variance of monthly occupancy rates by region and year. These values were then matched to the hotels according to their location and reporting year.

Control Variables – Orbis

We utilized Orbis database to obtain comprehensive financial information about the Portuguese hotels included in our analysis from 2009-2015 (see [Table 11: Hotel's Financial variables](#) for details). The financial variables used in this study to characterize hotels include: Total assets, which represent the resources owned by the hotel (i.e., the sum of fixed and current assets), were transformed using the natural logarithm and Winsorized at a 5% significance level (f_ln_total_assets_w), the debt ratio (f_debt_r), calculated as the ratio of total liabilities to shareholders' equity, the efficiency ratio (f_efficiency_r), defined as the ratio of income to total assets, the growth ratio (f_growth_r), which measures the ratio of hotel revenues in the current period relative to the previous period, the liquidity ratio (f_liquidity_r), computed as the ratio of current assets (excluding inventories) to current liabilities; and finally, earnings before interest and taxes margin (f_profit_r), which captures operational profitability.

Dependent variable

Orbis dataset was used to collect the financial variables to calculate the Z-Score (f_zscore_w) [2], which serve as the dependent variable in this study (see [Annex 04 – Z-Score](#) for details). This measure is used to assess the hotels' credit risk (Vivel-Búa et al., 2018). The Z-Score formula incorporates three key components: profitability (f_roaa_w , ROAA, μ), capital structure ($f_equitycapital_total_assets_w$ k), and profit volatility ($f_roaa_volatility_w$ σ). Profitability, measured by ROAA, reflects how efficiently each hotel utilizes its assets to generate earnings, with higher values indicating better performance. Capital structure, represented by the capital ratio (k), assesses the proportion of total assets financed by equity rather than debt, where a higher ratio indicates stronger financial health and a reduced reliance on external borrowing. Finally, profit volatility (σ) captures fluctuations in ROAA over time, with greater volatility higher risk. Inconsistent profits can hinder an enterprise's ability to meet financial obligations, thereby increasing credit risk. By combining these elements, the Z-Score provides a comprehensive evaluation of credit risk, enabling the identification of hotels with elevated risks of financial distress.

$$Z - score = \frac{\mu + k}{\sigma} \quad (2)$$

μ = return on average assets (ROAA), k = capital ratio (capital level/total assets), σ = standard deviation (ROAA volatility).

Descriptive analysis

Summary statistics for the location and financial variables are presented in [Table 1 Descriptives](#), providing key insights into hotel performance and credit risk. Financial data are reported annually from 2010 to 2015, allowing for temporal analysis. [Annex 05– Data -Descriptives](#) outlines the main data preparation procedures, while [Annex 07 - Descriptive - Missing Values of Hotels Characteristics and Financial Variables](#) and [Annex 08 - Descriptive Analysis of Financial Hotel Variables - with Years](#) present information on missing values and annual breakdowns.

The analysis of correlation coefficients between the variable (f_zscore_w) and the location-based distance indicators exhibit, as expected, negative correlations. This shows that the lower the distance from the hotels' location from the touristic location, the highest the Z-Score (i.e. lower credit risk). Similar patterns are observed for distances from types of touristic points, namely beaches, castles and churches (see [Annex 06 - Z-Score Correlations on Distance and Financial Variable](#), for details).

The correlation analysis between (f_zscore_w) and explanatory variables shows mostly weak to moderate relationships. Operational factors like room occupancy and event facilities show slight increases in risk with higher usage. Hotel quality ratings negatively correlate with the Z-score, suggesting higher-rated hotels may face greater risk. Overall, both financial and location factors help explain hotel credit risk in Portugal (see [Annex 06 - Z-Score Correlations on Distance and Financial Variable](#) , for details).

Table 1 - Descriptives

Variables	N	Mean	Std. dev.	Min	Max
f_debt_r	1,940	0.92	4.41	-56.56	124.97
f_efficiency_r	1,916	0	0.23	-3.44	5.58
f_growth_r	995	1,311.73	29,229.28	-1.21	796,338.10
f_liquidity_r	1,913	8.97	184.73	0	7,972.70
f_ln_total_assets_w	1,930	14.21	1.74	7.82	16.15
f_mkt_share_conc_nutsii	6,378	0.37	0.34	0	1
f_mkt_share_conc_nutsiii	6,378	0.35	0.34	0	1
f_mkt_share_nutsii	1,893	0.03	0.11	-0.02	1
f_mkt_share_nutsiii	1,893	0.07	0.17	-0.29	1
f_profit_r	1,828	2.60	21.38	-97.86	97.97
f_zscore_w	1,930	0.33	0.87	-2.29	20.90
ht_events_meetings	7,441	107.17	229.00	0	3,000
ht_location_1	7,441	3.44	5.78	0.04	44.40
ht_location_2	7,441	5.65	7.83	0.08	53.64
ht_location_3	7,441	8.26	9.48	0.24	58.58
ht_location_4	7,441	10.93	10.90	0.55	67.78
ht_location_5	7,441	32.42	73.37	0.79	308.96
ht_location_beach1	7,441	14.02	14.75	0.05	73.83
ht_location_beach2	7,441	17.87	16.81	0.24	83.50
ht_location_beach3	7,441	22.37	20.33	0.35	96.53
ht_location_beach4	7,441	34.47	37.63	0.64	162.22
ht_location_beach5	7,441	53.18	87.90	1.26	368.43
ht_location_castle1	7,441	87.51	259.70	0.11	1,037.65
ht_location_castle2	7,441	103.71	259.92	7.47	1,044.10
ht_location_castle3	7,441	144.69	311.56	15.82	1,177.70
ht_location_castle4	7,441	167.39	341.08	25.98	1,512.19
ht_location_castle5	7,441	183.38	357.41	31.78	1,684.60
ht_location_church1	7,441	7.02	10.85	0.04	80.34
ht_location_church2	7,441	56.50	171.59	0.08	680.93
ht_location_church3	7,441	73.05	215.26	0.26	854.76
ht_location_church4	7,441	81.57	230.43	0.61	922.07
ht_location_church5	7,441	87.84	238.20	0.85	958.92
ht_quality_stars	7,441	3.30	0.98	1	5.00
ht_room_ocup_nutsii	6,378	0.50	0.11	0.34	0.75
ht_room_ocup_nutsiii	6,378	0.49	0.13	0.22	0.75
ht_room_ocup_sea_nutsii	6,378	0.02	0.01	0.01	0.06
ht_room_ocup_sea_nutsiii	6,222	0.03	0.01	0.00	0.06

Note: N denotes number of observations. Std. dev denotes standard deviation. Min denotes minimum. Max denotes maximum.

3.1 METHODOLOGY

Dynamic panel data

Previous studies have employed pooled Ordinary Least Squares (OLS), probit, and logit models to examine the determinants of credit risk. This study uses the Z-score as the dependent variable and applies a panel data methodology, which combines cross-sectional and time-series data by observing multiple hotel characteristics over time horizon of five years. This approach enhances the robustness of the analysis by accounting for unobserved heterogeneity and capturing potential dynamics in the relationship between explanatory variables and credit risk. To address endogeneity concerns, the one-step System Generalized Method of Moments (System GMM) estimator is employed, which is well-suited for dynamic panel data where the Z-Score dependent variable depends on its past values. Finally, the Arellano-Bond test for autocorrelation is conducted to validate the model, ensuring that residuals do not exhibit second-order serial correlation and that the moment conditions underlying the GMM estimator are satisfied.

We incorporated these five distance-based variables as the main independent variables in our analysis of hotels' creditworthiness.

4. RESULTS AND DISCUSSION

This study provides empirical evidence on the relationship between individual hotel credit risk and distances from hotel location to touristic places. The findings consistently indicate that the proximity of the hotels to tourist attractions, for instance beaches, castles, and churches has a significant impact on the reduction of credit risk. Hotels located closer to these touristic amenities tend to exhibit lower financial risk.

Among operational variables, hotels with greater event and meeting capacity, higher room occupancy rates, and stronger operational efficiency were systematically associated with more favorable credit risk profiles. These results highlight the role of effective hotel management and service capacity in mitigating financial vulnerability.

Conversely, financial indicators such as total assets, liquidity, profitability, and growth were negatively associated with the Z-score, suggesting that higher values of these variables correspond to greater financial vulnerability. This finding highlights the potential for future research to explore how interactions between location and financial factors can better explain variations in the Z-score and overall risk.

Several financial control variables were included to capture differences related to hotel location, as well as firm-level characteristics such as size, profitability, market position, and capital structure. Some of these variables exhibited significant associations with credit risk; however, a more detailed examination of their individual effects falls beyond the scope of this study and is recommended for future research.

While all three types of tourist locations contributed to explaining credit risk (i.e. beaches, churches and castles), the significance and magnitude of the relationships varied. Proximity to castles emerged as the most reliable location-based predictor of lower credit risk across all models. Proximity to beaches also showed a consistent effect, while the influence of churches, though present, was comparatively less robust.

Overall, the results suggest that both physical location and internal financial and operational characteristics play a central role in determining hotel credit risk. The integration of location-based metrics into credit risk assessment models provides a more comprehensive view of hotel vulnerability and resilience. These insights are particularly relevant for investors, financial institutions, and policymakers focused on the tourism and hospitality sector in Portugal.

Table 2: Determinants of credit risk in the Portuguese hotel sector by different definitions of touristic points

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.584 (0.115) ***	1.588 (0.113) ***	1.586 (0.111) ***	1.504 (0.132) ***	1.457 (0.117) ***
ht_location_1	0.027 (0.045)	-	-	-	-
ht_location_2	-	0.031 (0.045)	-	-	-
ht_location_3	-	-	0.032 (0.046)	-	-
ht_location_4	-	-	-	-0.047 (0.038)	-
ht_location_5	-	-	-	-	-0.010 (0.002) ***
f_mkt_share_conc_nutsii	0.149 (0.887)	0.092 (0.878)	0.030 (0.884)	0.357 (1.079)	0.134 (0.959)
ht_room_ocup_sea_nutsii	28.273 (28.280)	34.397 (31.233)	38.907 (34.523)	2.256 (36.451)	-19.589 (31.453)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) *	0.001 (0.001) **
f_ln_total_assets_w	-0.293 (0.170) *	-0.298 (0.162) *	-0.299 (0.160) *	-0.397 (0.191) **	-0.464 (0.170) ***
f_liquidity_r	-0.011 (0.005) **	-0.012 (0.005) **	-0.012 (0.006) **	-0.004 (0.005)	-0.009 (0.003) **
f_debt_r	0.056 (0.072)	0.053 (0.071)	0.057 (0.070)	0.045 (0.086)	0.036 (0.077)
f_efficiency_r	1.838 (0.428) ***	1.785 (0.450) ***	1.747 (0.478) ***	2.391 (0.533) ***	2.029 (0.378) ***
f_profit_r	-0.048 (0.014) ***	-0.047 (0.014) ***	-0.046 (0.014) ***	-0.066 (0.017) ***	-0.057 (0.012) ***
f_growth_r	-0.001 (0.001) *	-0.001 (0.001) *	-0.001 (0.001)	-0.001 (0.001) **	-0.001 (0.001) ***
ht_room_ocup_nutsii	5.170 (2.820) *	5.375 (2.849) *	5.355 (2.815) *	4.374 (3.239)	11.442 (3.468) ***
ht_quality_stars	0.171 (0.168)	0.153 (0.174)	0.146 (0.178)	0.321 (0.194) *	0.341 (0.162) **
f_mkt_share_nutsii	-0.005 (0.947)	0.026 (0.919)	0.081 (0.904)	0.579 (1.156)	1.760 (1.089)
2011.year	-1.104 (0.662) *	-1.072 (0.660)	-1.031 (0.671)	-1.436 (0.807) *	-1.261 (0.703) *
2012.year	-0.315 (0.419)	-0.251 (0.454)	-0.218 (0.478)	-0.845 (0.508) *	-0.669 (0.368) *
2013.year	0.081 (0.431)	0.152 (0.462)	0.200 (0.493)	-0.340 (0.556)	0.133 (0.440)
2014.year	-0.041 (0.162)	-0.027 (0.165)	-0.021 (0.168)	-0.154 (0.184)	0.183 (0.174)
Constant	-0.032 (2.507)	-0.272 (2.569)	-0.428 (2.675)	2.758 (2.708)	0.071 (2.062)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	926.590	948.820	958.580	645.220	804.650
AR(1)	-3.030 ***	-3.110 ***	-3.130 ***	-2.380 **	-2.540 ***
AR(2)	0.660	0.690	0.700	0.190	0.480
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

At the NUTS II level for distances including all touristic points, only ht_location_5—the average distance to beaches, castles, and churches—was statistically significant (coef. = -0.010 , p-value < 0.01), indicating that proximity to multiple tourist attractions is relevant in explaining credit risk. Among the control variables, ht_events_meetings (coef. ≈ 0.001 , p-value < 0.05), f_efficiency_r (coef. = 2.029 , p-value < 0.01), ht_room_ocup_nutsii (coef. = 11.442 , p-value < 0.01), and ht_quality_stars (coef. = 0.341 , p-value < 0.05) were positively associated with the Z-score, suggesting that hotels with more meeting facilities, higher efficiency, stronger occupancy, and higher classifications tend to show greater credit risk. In contrast, f_ln_total_assets_w (coef. = -0.464 , p-value < 0.01), f_liquidity_r (coef. = -0.009 , p-value < 0.05), and f_profit_r (coef. = -0.057 , p-

value < 0.01) were negatively related to Z-Score (i.e. higher credit risk), implying that hotels with smaller size, lower liquidity, and weaker profitability are less exposed to financial distress. Model robustness is confirmed with a strong Wald χ^2 statistic (804.65), significant first-order serial correlation (AR(1) = -2.54 , p-value < 0.01), and no evidence of second-order autocorrelation (AR(2) = 0.48 , p-value > 0.1).

Table 3: Determinants of Credit Risk in Portugal's NUTS II Regions: Beaches Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.603 (0.099) ***	1.591 (0.100) ***	1.569 (0.104) ***	1.472 (0.133) ***	1.462 (0.122) ***
ht_location_beach1	0.040 (0.017) **	-	-	-	-
ht_location_beach2	-	0.032 (0.015) **	-	-	-
ht_location_beach3	-	-	0.013 (0.013)	-	-
ht_location_beach4	-	-	-	-0.018 (0.006) ***	-
ht_location_beach5	-	-	-	-	-0.008 (0.002) ***
f_mkt_share_conc_nutsii	0.475 (0.839)	0.329 (0.840)	0.167 (0.876)	0.099 (1.094)	0.105 (1.004)
ht_room_ocup_sea_nutsii	64.406 (30.844) **	40.714 (26.732)	23.469 (26.588)	9.631 (33.556)	-2.341 (31.372)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) ***	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **
f_ln_total_assets_w	-0.315 (0.143) **	-0.322 (0.145) **	-0.326 (0.152) **	-0.446 (0.194) **	-0.458 (0.178) **
f_liquidity_r	-0.013 (0.003) ***	-0.011 (0.003) ***	-0.010 (0.003) ***	-0.007 (0.004) *	-0.009 (0.004) **
f_debt_r	0.115 (0.070)	0.136 (0.076) *	0.110 (0.086)	-0.049 (0.096)	-0.010 (0.083)
f_efficiency_r	1.447 (0.400) ***	1.454 (0.417) ***	1.715 (0.441) ***	2.364 (0.452) ***	2.134 (0.398) ***
f_profit_r	-0.035 (0.013) ***	-0.036 (0.013) ***	-0.044 (0.014) ***	-0.070 (0.015) ***	-0.062 (0.013) ***
f_growth_r	-0.001 (0.001) *	-0.001 (0.001) *	-0.001 (0.001) **	-0.001 (0.001) ***	-0.001 (0.001) ***
ht_room_ocup_nutsii	5.172 (2.516) **	5.918 (2.605) **	5.426 (2.768) *	8.687 (3.626) **	10.820 (3.538) ***
ht_quality_stars	0.160 (0.139)	0.177 (0.140)	0.204 (0.146)	0.299 (0.183)	0.311 (0.168) *
f_mkt_share_nutsii	0.006 (0.854)	-0.134 (0.872)	-0.112 (0.939)	1.930 (1.304)	1.997 (1.171) *
2011.year	-1.164 (0.605) *	-1.127 (0.612) *	-1.107 (0.646) *	-1.312 (0.803)	-1.258 (0.736) *
2012.year	-0.141 (0.343)	-0.208 (0.340)	-0.340 (0.355)	-0.693 (0.423)	-0.628 (0.383)
2013.year	0.000 (0.377)	0.085 (0.384)	0.077 (0.409)	0.015 (0.500)	0.134 (0.461)
2014.year	0.026 (0.144)	0.058 (0.153)	-0.002 (0.166)	0.008 (0.183)	0.135 (0.177)
Constant	-1.264 (2.007)	-1.023 (2.021)	0.024 (2.090)	1.169 (2.337)	0.161 (2.156)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	1074.060	1050.240	950.520	616.740	734.520
AR(1)	-3.010 ***	-2.920 ***	-2.890 ***	-2.370 **	-2.500 **
AR(2)	0.820	0.870	0.760	-0.010	-0.260
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

At the NUTS II level, all beach distance variables were statistically significant except ht_location_beach3 (p-value > 0.1), indicating that proximity to beaches plays a consistent role in explaining hotel credit risk. Regarding the control variables in the most robust regression model featuring ht_location_beach5, with ht_events_meetings (coef. ≈ 0.001 , p-value < 0.05), f_efficiency_r (coef. = 2.134 , p-value < 0.01), and ht_room_ocup_nutsii (coef. = 10.820 , p-value < 0.01) were positively associated with the Z-score, suggesting that hotels with more meeting rooms, higher efficiency, and higher occupancy rates tend to have lower credit risk. Conversely, f_ln_total_assets_w (coef. = -0.458 , p-value < 0.05), f_liquidity_r (coef. = -0.009 , p-value < 0.05), f_profit_r

(coef. = -0.062 , p-value < 0.01), and f_growth_r (coef. = -0.001 , p-value < 0.01) had significant negative effects on Z-Score, indicating that higher asset levels, liquidity, profitability, and growth are linked to lower Z-Score, i.e. greater financial vulnerability. Model diagnostics confirm robustness with a strong Wald χ^2 (734.52), significant AR(1) (-2.50 , p-value < 0.05), and no second-order autocorrelation (AR(2) = -0.26 , p-value > 0.1).

Table 4: Determinants of Credit Risk in Portugal's NUTS II Regions: Castles Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.466 (0.112) ***	1.471 (0.112) ***	1.507 (0.115) ***	1.517 (0.116) ***	1.521 (0.116) ***
ht_location_castle1	-0.002 (0.000) ***	-	-	-	-
ht_location_castle2	-	-0.002 (0.000) ***	-	-	-
ht_location_castle3	-	-	-0.001 (0.000) ***	-	-
ht_location_castle4	-	-	-	-0.001 (0.000) ***	-
ht_location_castle5	-	-	-	-	-0.001 (0.000) ***
f_mkt_share_conc_nutsii	0.102 (0.927)	0.116 (0.927)	0.213 (0.965)	0.229 (0.974)	0.236 (0.978)
ht_room_ocup_sea_nutsii	-13.183 (29.799)	-8.898 (29.434)	29.509 (29.362)	37.444 (29.943)	40.839 (30.272)
ht_events_meetings	0.001 (0.001) ***	0.001 (0.001) ***	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **
f_ln_total_assets_w	-0.451 (0.164) ***	-0.440 (0.163) ***	-0.380 (0.168) **	-0.364 (0.169) **	-0.358 (0.169) **
f_liquidity_r	-0.010 (0.003) ***	-0.010 (0.003) ***	-0.011 (0.004) ***	-0.011 (0.004) ***	-0.011 (0.004) ***
f_debt_r	0.036 (0.074)	0.025 (0.075)	-0.042 (0.084)	-0.052 (0.087)	-0.055 (0.088)
f_efficiency_r	1.923 (0.366) ***	1.922 (0.365) ***	1.980 (0.380) ***	1.992 (0.384) ***	1.996 (0.385) ***
f_profit_r	-0.054 (0.011) ***	-0.055 (0.011) ***	-0.058 (0.012) ***	-0.059 (0.012) ***	-0.059 (0.012) ***
f_growth_r	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***
ht_room_ocup_nutsii	11.703 (3.390) ***	11.494 (3.362) ***	9.314 (3.270) ***	8.679 (3.242) ***	8.406 (3.229) ***
ht_quality_stars	0.313 (0.156) **	0.300 (0.155) *	0.208 (0.160)	0.188 (0.162)	0.180 (0.162)
f_mkt_share_nutsii	1.729 (1.049) *	1.736 (1.050) *	1.687 (1.109)	1.610 (1.119)	1.571 (1.122)
2011.year	-1.199 (0.680) *	-1.196 (0.679) *	-1.175 (0.707) *	-1.169 (0.714)	-1.166 (0.716)
2012.year	-0.571 (0.353)	-0.552 (0.352)	-0.403 (0.366)	-0.369 (0.370)	-0.355 (0.372)
2013.year	0.208 (0.428)	0.198 (0.427)	0.114 (0.443)	0.095 (0.446)	0.087 (0.448)
2014.year	0.205 (0.171)	0.192 (0.170)	0.069 (0.166)	0.039 (0.165)	0.025 (0.165)
Constant	-0.429 (2.014)	-0.491 (2.016)	-0.773 (2.129)	-0.775 (2.156)	-0.761 (2.166)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	860.650	861.830	792.310	776.950	771.310
AR(1)	-2.640 ***	-2.660 ***	-2.690 ***	-2.690 ***	-2.700 ***
AR(2)	0.570	0.550	0.280	0.230	0.220
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

At the NUTS II level, when the distance to castles was assessed individually, all five distances (ht_location_castle1 to ht_location_castle5) were statistically significant at the 1% level, each showing a negative relationship with the Z-score (i.e. lower credit risk). This indicates that greater proximity to castles is associated with a decrease in credit risk. Among the control variables, ht_events_meetings, f_efficiency_r, and ht_room_ocup_nutsii remained positively associated with the Z-score across all specifications), suggesting that hotels with more events, higher efficiency, and better occupancy are more likely to a higher credit risk. On the other hand, f_ln_total_assets_w,

f_liquidity_r, f_profit_r, and f_growth_r were consistently negatively related to the Z-score. indicating that larger hotels show lower Z-Score, i.e. higher risk.

Table 5: Determinants of Credit Risk in Portugal's NUTS II Regions: Churches Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.557 (0.113) ***	1.461 (0.114) ***	1.462 (0.113) ***	1.463 (0.113) ***	1.464 (0.113) ***
ht_location_church1	-0.012 (0.042)	-	-	-	-
ht_location_church2	-	-0.004 (0.001) ***	-	-	-
ht_location_church3	-	-	-0.003 (0.000) ***	-	-
ht_location_church4	-	-	-	-0.003 (0.000) ***	-
ht_location_church5	-	-	-	-	-0.003 (0.000) ***
f_mkt_share_conc_nutsii	0.190 (0.980)	0.099 (0.937)	0.100 (0.936)	0.102 (0.936)	0.107 (0.936)
ht_room_ocup_sea_nutsii	21.017 (30.035)	-21.307 (30.852)	-20.069 (30.694)	-19.500 (30.650)	-18.783 (30.577)
ht_events_meetings	0.001 (0.001) *	0.001 (0.001) ***	0.001 (0.001) ***	0.001 (0.001) ***	0.001 (0.001) ***
f_ln_total_assets_w	-0.360 (0.185) *	-0.466 (0.166) ***	-0.462 (0.166) ***	-0.458 (0.166) ***	-0.455 (0.165) ***
f_liquidity_r	-0.007 (0.006)	-0.009 (0.003) **	-0.009 (0.003) **	-0.009 (0.003) **	-0.009 (0.003) **
f_debt_r	0.072 (0.083)	0.048 (0.075)	0.046 (0.075)	0.044 (0.075)	0.043 (0.075)
f_efficiency_r	2.066 (0.469) ***	1.959 (0.369) ***	1.955 (0.369) ***	1.956 (0.369) ***	1.955 (0.369) ***
f_profit_r	-0.055 (0.013) ***	-0.055 (0.012) ***	-0.055 (0.012) ***	-0.055 (0.012) ***	-0.055 (0.012) ***
f_growth_r	-0.001 (0.001) **	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***
ht_room_ocup_nutsii	3.919 (3.859)	11.867 (3.437) ***	11.843 (3.431) ***	11.729 (3.416) ***	11.615 (3.401) ***
ht_quality_stars	0.272 (0.237)	0.346 (0.159) **	0.338 (0.158) **	0.335 (0.158) **	0.333 (0.158) **
f_mkt_share_nutsii	0.147 (0.993)	1.735 (1.059)	1.738 (1.058)	1.736 (1.058)	1.729 (1.058)
2011.year	-1.286 (0.793)	-1.233 (0.687) *	-1.227 (0.686) *	-1.237 (0.687) *	-1.245 (0.686) *
2012.year	-0.594 (0.568)	-0.637 (0.358) *	-0.627 (0.357) *	-0.634 (0.358) *	-0.639 (0.358) *
2013.year	-0.121 (0.585)	0.190 (0.432)	0.193 (0.431)	0.183 (0.431)	0.174 (0.431)
2014.year	-0.118 (0.199)	0.215 (0.174)	0.214 (0.173)	0.207 (0.173)	0.201 (0.172)
Constant	1.671 (3.201)	-0.198 (2.025)	-0.255 (2.025)	-0.236 (2.024)	-0.216 (2.023)
Observations (no. of hotels)	960	960	960	960	960
Wald (χ^2)	789.330	842.660	844.700	844.460	845.160
AR(1)	-2.670 ***	-2.580 **	-2.590 **	-2.600 ***	-2.600 ***
AR(2)	0.460	0.570	0.560	0.560	0.560
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Regarding the control variables in the most robust regression model with ht_location_church5, ht_events_meetings, f_efficiency_r, ht_room_ocup_nutsii, and f_profit_r demonstrated significant associations with the Z-score. Specifically, the presence of more meeting rooms, greater operational efficiency, and higher occupancy rates were linked to higher Z-Scores (i.e. improved credit risk profiles). Conversely, larger total assets and liquidity ratios exhibited a negative relationship with the Z-score (i.e. higher risk), indicating larger hotels show lower Z-Score, i.e. higher risk. Additionally, profitability and growth rates contributed negatively, further emphasizing their importance in assessing hotel credit risk. Other variables such as market share concentration and debt ratios showed no consistent significance. The tests for serial correlation (AR(1)) confirmed model validity, while the Wald statistics indicated strong joint significance of the explanatory variables.

Table 6: Determinants of Credit Risk in Portugal's NUTS III Regions: All Location Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.571 (0.136) ***	1.574 (0.136) ***	1.589 (0.133) ***	1.450 (0.173) ***	1.439 (0.151) ***
ht_location_1	0.007 (0.048)	-	-	-	-
ht_location_2	-	0.009 (0.049)	-	-	-
ht_location_3	-	-	0.022 (0.053)	-	-
ht_location_4	-	-	-	-0.075 (0.052)	-
ht_location_5	-	-	-	-	-0.010 (0.003) ***
f_mkt_share_conc_nutsiii	-0.039 (1.222)	-0.090 (1.246)	-0.265 (1.301)	0.786 (1.665)	-0.849 (1.490)
ht_room_ocup_sea_nutsiii	26.836 (27.849)	28.740 (30.975)	35.148 (34.275)	-7.044 (41.061)	-29.911 (37.240)
ht_events_meetings	0.001 (0.001) *	0.001 (0.001) *	0.001 (0.001) **	0.001 (0.001) *	0.001 (0.001) **
f_ln_total_assets_w	-0.343 (0.194) *	-0.339 (0.192) *	-0.313 (0.192)	-0.529 (0.244) **	-0.554 (0.210) ***
f_liquidity_r	-0.005 (0.005)	-0.005 (0.005)	-0.006 (0.006)	0.001 (0.006)	-0.002 (0.004)
f_debt_r	0.062 (0.083)	0.059 (0.085)	0.054 (0.083)	0.049 (0.106)	0.022 (0.100)
f_efficiency_r	2.140 (0.522) ***	2.119 (0.542) ***	2.027 (0.561) ***	2.886 (0.747) ***	2.672 (0.552) ***
f_profit_r	-0.060 (0.018) ***	-0.059 (0.018) ***	-0.056 (0.018) ***	-0.086 (0.025) ***	-0.079 (0.018) ***
f_growth_r	-0.001 (0.001) *	-0.001 (0.001) *	-0.001 (0.001)	-0.001 (0.001) **	-0.001 (0.001) ***
ht_room_ocup_nutsiii	5.706 (2.788) **	5.656 (2.795) **	5.527 (2.732) **	7.469 (3.752) **	12.096 (3.985) ***
ht_quality_stars	0.173 (0.178)	0.165 (0.188)	0.133 (0.199)	0.364 (0.242)	0.380 (0.205) *
f_mkt_share_nutsiii	1.220 (1.066)	1.221 (1.057)	1.230 (1.027)	1.641 (1.388)	2.804 (1.383) **
2011.year	-1.828 (0.636) ***	-1.811 (0.643) ***	-1.732 (0.661) ***	-2.382 (0.893) ***	-2.004 (0.759) ***
2012.year	-0.932 (0.456) **	-0.902 (0.511) *	-0.785 (0.579)	-1.713 (0.747) **	-1.350 (0.514) ***
2013.year	-0.047 (0.309)	-0.028 (0.342)	-0.047 (0.385)	-0.521 (0.488)	0.088 (0.348)
2014.year	-0.068 (0.167)	-0.068 (0.165)	-0.059 (0.162)	-0.093 (0.211)	0.220 (0.221)
Constant	0.558 (2.482)	0.478 (2.512)	0.002 (2.667)	3.369 (3.148)	1.433 (2.416)
Observations (no. of hotels)	941	941	941	941	941
Wald (χ^2)	709.930	719.300	760.260	436.880	503.750
AR(1)	-2.730 ***	-2.810 ***	-2.920 ***	-2.050 **	-2.090 **
AR(2)	0.120	0.130	0.200	-0.340	-0.060
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Regarding the control variables in the most robust regression model at the NUTS III regional level, ht_location_5, ht_events_meetings, f_efficiency_r, ht_room_ocup_nutsiii, and f_profit_r showed significant relationships with the Z-score. The number of meeting and event rooms, operational efficiency, and room occupancy rates were positively associated with improved credit risk profiles. Conversely, total assets showed a negative association, suggesting larger hotels show lower Z-Score, i.e. higher risk. Profitability and growth rates consistently contributed negatively to the Z-score, indicating their relevance for risk assessment. Market share concentration, liquidity, and debt ratios did not present consistent significance. The AR(1) test confirmed absence of first-order serial

correlation, supporting model validity, and the Wald test indicated overall joint significance of the included variables.

Table 7: Determinants of Credit Risk in Portugal's NUTS III Regions: Beaches Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.604 (0.116) ***	1.602 (0.115) ***	1.576 (0.118) ***	1.485 (0.150) ***	1.469 (0.148) ***
ht_location_beach1	0.034 (0.017) *	-	-	-	-
ht_location_beach2	-	0.034 (0.018) *	-	-	-
ht_location_beach3	-	-	0.013 (0.017)	-	-
ht_location_beach4	-	-	-	-0.014 (0.006) **	-
ht_location_beach5	-	-	-	-	-0.006 (0.002) ***
f_mkt_share_conc_nutsiii	-0.443 (1.168)	-0.641 (1.185)	-0.304 (1.229)	-0.312 (1.482)	-0.662 (1.471)
ht_room_ocup_sea_nutsiii	43.461 (26.665)	28.086 (24.868)	20.484 (26.592)	12.055 (32.783)	-4.294 (33.604)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) *	0.001 (0.001) *
f_ln_total_assets_w	-0.301 (0.159) *	-0.314 (0.157) **	-0.341 (0.162) **	-0.475 (0.208) **	-0.502 (0.205) **
f_liquidity_r	-0.006 (0.003)	-0.005 (0.003)	-0.005 (0.003)	-0.002 (0.004)	-0.002 (0.004)
f_debt_r	0.097 (0.079)	0.136 (0.086)	0.115 (0.103)	-0.046 (0.111)	-0.020 (0.103)
f_efficiency_r	1.883 (0.439) ***	1.804 (0.457) ***	1.980 (0.494) ***	2.670 (0.572) ***	2.631 (0.548) ***
f_profit_r	-0.051 (0.014) ***	-0.048 (0.015) ***	-0.054 (0.017) ***	-0.080 (0.019) ***	-0.078 (0.018) ***
f_growth_r	-0.001 (0.001) **	-0.001 (0.001) **	-0.001 (0.001) **	-0.001 (0.001) **	-0.001 (0.001) ***
ht_room_ocup_nutsiii	6.258 (2.626) **	6.746 (2.646) **	6.385 (2.807) **	8.389 (3.558) **	10.181 (3.696) ***
ht_quality_stars	0.107 (0.157)	0.128 (0.153)	0.168 (0.158)	0.286 (0.200)	0.316 (0.198)
f_mkt_share_nutsiii	1.285 (1.000)	1.298 (0.990)	1.208 (1.028)	2.275 (1.365) *	2.570 (1.358) *
2011.year	-1.613 (0.605) ***	-1.499 (0.617) **	-1.638 (0.661) **	-2.168 (0.778) ***	-2.063 (0.755) ***
2012.year	-0.684 (0.414) *	-0.648 (0.422)	-0.801 (0.448) *	-1.312 (0.524) **	-1.279 (0.506) **
2013.year	0.071 (0.279)	0.178 (0.299)	0.083 (0.335)	-0.136 (0.349)	0.004 (0.343)
2014.year	0.026 (0.163)	0.083 (0.175)	0.017 (0.195)	-0.019 (0.200)	0.102 (0.206)
Constant	-1.080 (2.112)	-0.982 (2.103)	0.000 (2.172)	1.739 (2.466)	1.338 (2.397)
Observations (no. of hotels)	941	941	941	941	941
Wald (χ^2)	806.440	821.470	760.210	490.990	509.860
AR(1)	-2.830 ***	-2.770 ***	-2.660 ***	-2.220 **	-2.200 **
AR(2)	0.370	0.470	0.300	-0.290	-0.160
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Regarding the control variables in the most robust regression model at the NUTS III regional level, ht_location_beach5, ht_events_meetings, f_efficiency_r, ht_room_ocup_nutsiii, and f_profit_r showed significant relationships with the Z-score. The number of meeting and event rooms, operational efficiency, and room occupancy rates were positively associated with improved credit risk profiles. Conversely, total assets showed a negative association, suggesting that larger hotels show lower Z-Score, i.e. higher risk. Profitability and growth

rates consistently contributed negatively to the Z-score, indicating their relevance for risk assessment. Market share concentration, liquidity, and debt ratios did not present consistent significance. The AR(1) test confirmed absence of first-order serial correlation, supporting model validity, and the Wald test indicated overall joint significance of the included variables.

Table 8: Determinants of Credit Risk in Portugal's NUTS III Regions: Castles Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.452 (0.146) ***	1.460 (0.144) ***	1.511 (0.139) ***	1.522 (0.137) ***	1.526 (0.137) ***
ht_location_castle1	-0.002 (0.000) ***	-	-	-	-
ht_location_castle2	-	-0.002 (0.000) ***	-	-	-
ht_location_castle3	-	-	-0.001 (0.000) **	-	-
ht_location_castle4	-	-	-	-0.001 (0.000) **	-
ht_location_castle5	-	-	-	-	-0.001 (0.000) **
f_mkt_share_conc_nutsiii	-0.968 (1.459)	-0.897 (1.445)	-0.433 (1.394)	-0.342 (1.382)	-0.306 (1.377)
ht_room_ocup_sea_nutsiii	-24.291 (35.350)	-18.790 (34.391)	20.586 (30.273)	27.429 (30.034)	30.253 (29.994)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) *	0.001 (0.001) *	0.001 (0.001) *
f_ln_total_assets_w	-0.529 (0.202) ***	-0.510 (0.200) **	-0.405 (0.189) **	-0.385 (0.188) **	-0.378 (0.187) **
f_liquidity_r	-0.002 (0.004)	-0.002 (0.004)	-0.003 (0.004)	-0.004 (0.004)	-0.004 (0.004)
f_debt_r	0.018 (0.097)	0.007 (0.097)	-0.044 (0.103)	-0.049 (0.104)	-0.051 (0.105)
f_efficiency_r	2.576 (0.528) ***	2.553 (0.523) ***	2.432 (0.507) ***	2.404 (0.502) ***	2.391 (0.500) ***
f_profit_r	-0.076 (0.017) ***	-0.076 (0.017) ***	-0.073 (0.017) ***	-0.072 (0.017) ***	-0.072 (0.017) ***
f_growth_r	-0.001 (0.001) ***	-0.001 (0.001) **	-0.001 (0.001) **	-0.001 (0.001) **	-0.001 (0.001) **
ht_room_ocup_nutsiii	12.097 (3.884) ***	11.718 (3.809) ***	8.969 (3.399) ***	8.423 (3.322) **	8.188 (3.289) **
ht_quality_stars	0.345 (0.196) *	0.325 (0.193) *	0.202 (0.183)	0.180 (0.182)	0.172 (0.181)
f_mkt_share_nutsiii	2.755 (1.341) **	2.690 (1.327) **	2.169 (1.260) *	2.050 (1.244) *	2.000 (1.237)
2011.year	-1.917 (0.737) ***	-1.931 (0.732) ***	-1.994 (0.719) ***	-1.999 (0.715) ***	-2.000 (0.713) ***
2012.year	-1.229 (0.491) **	-1.214 (0.487) **	-1.085 (0.472) **	-1.056 (0.468) **	-1.044 (0.466) *
2013.year	0.158 (0.344)	0.134 (0.340)	-0.014 (0.326)	-0.038 (0.324)	-0.048 (0.323)
2014.year	0.229 (0.217)	0.204 (0.213)	0.029 (0.191)	-0.002 (0.188)	-0.015 (0.186)
Constant	0.958 (2.343)	0.840 (2.327)	0.170 (2.289)	0.065 (2.282)	0.035 (2.277)
Observations (no. of hotels)	941	941	941	941	941
Wald (χ^2)	531.150	537.820	559.340	565.440	568.740
AR(1)	-2.160 **	-2.190 **	-2.380 **	-2.410 **	-2.430 **
AR(2)	-0.010	-0.030	-0.160	-0.170	-0.180
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Regarding the control variables in the most robust regression model at the NUTS III regional, ht_location_castle5, ht_events_meetings, f_efficiency_r, ht_room_ocup_nutsiii, and f_profit_r showed significant relationships with the Z-score. The number of meeting and event rooms, operational efficiency, and room occupancy rates were positively associated with improved credit risk profiles. Conversely, total assets showed a negative association,

suggesting that larger hotels show lower Z-Score, i.e. higher risk. Profitability and growth rates consistently contributed negatively to the Z-score, indicating their relevance for risk assessment. Market share concentration, liquidity, and debt ratios did not present consistent significance. The AR(1) test confirmed absence of first-order serial correlation, supporting model validity, and the Wald test indicated overall joint significance of the included variables.

Table 9: Determinants of Credit Risk in Portugal's NUTS III Regions: Churches Distances

Variables	Dynamic Panel Data				
	Dependent Variable:				
	Z-score				
	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)	Coef (Std. error)
L.Z-score_w	1.562 (0.122) ***	1.445 (0.148) ***	1.447 (0.148) ***	1.448 (0.147) ***	1.449 (0.147) ***
ht_location_church1	-0.004 (0.037)	-	-	-	-
ht_location_church2	-	-0.004 (0.001) ***	-	-	-
ht_location_church3	-	-	-0.003 (0.001) ***	-	-
ht_location_church4	-	-	-	-0.003 (0.001) ***	-
ht_location_church5	-	-	-	-	-0.003 (0.001) ***
f_mkt_share_conc_nutsiii	-0.041 (1.231)	-1.019 (1.478)	-1.006 (1.474)	-1.008 (1.474)	-1.002 (1.473)
ht_room_ocup_sea_nutsiii	26.647 (28.074)	-33.019 (37.090)	-31.378 (36.751)	-30.851 (36.680)	-30.305 (36.589)
ht_events_meetings	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **	0.001 (0.001) **
f_ln_total_assets_w	-0.348 (0.188) *	-0.556 (0.207) ***	-0.549 (0.206) ***	-0.545 (0.206) ***	-0.541 (0.205) ***
f_liquidity_r	-0.005 (0.005)	-0.002 (0.004)	-0.002 (0.004)	-0.002 (0.004)	-0.002 (0.004)
f_debt_r	0.060 (0.088)	0.033 (0.098)	0.030 (0.097)	0.028 (0.098)	0.027 (0.098)
f_efficiency_r	2.152 (0.504) ***	2.628 (0.538) ***	2.618 (0.536) ***	2.617 (0.536) ***	2.615 (0.536) ***
f_profit_r	-0.061 (0.016) ***	-0.077 (0.018) ***	-0.077 (0.018) ***	-0.077 (0.018) ***	-0.077 (0.018) ***
f_growth_r	-0.001 (0.001) **	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***	-0.001 (0.001) ***
ht_room_ocup_nutsiii	5.858 (2.970) **	12.352 (3.957) ***	12.273 (3.937) ***	12.192 (3.925) ***	12.132 (3.914) ***
ht_quality_stars	0.168 (0.211)	0.390 (0.203) *	0.379 (0.201) *	0.375 (0.201) *	0.371 (0.200) *
f_mkt_share_nutsiii	1.231 (1.070)	2.860 (1.367) **	2.843 (1.362) **	2.834 (1.361) **	2.821 (1.359) **
2011.year	-1.825 (0.645) ***	-1.938 (0.746) ***	-1.938 (0.744) ***	-1.940 (0.744) ***	-1.941 (0.744) ***
2012.year	-0.927 (0.500) *	-1.301 (0.501) **	-1.291 (0.500) **	-1.292 (0.500) **	-1.293 (0.500) **
2013.year	-0.039 (0.358)	0.154 (0.347)	0.152 (0.346)	0.146 (0.346)	0.140 (0.345)
2014.year	-0.066 (0.173)	0.252 (0.222)	0.247 (0.221)	0.241 (0.220)	0.236 (0.220)
Constant	0.559 (2.685)	1.292 (2.375)	1.221 (2.368)	1.220 (2.368)	1.214 (2.366)
Observations (no. of hotels)	941	941	941	941	941
Wald (χ^2)	700.160	519.750	522.120	522.170	522.540
AR(1)	-2.680 ***	-2.100 **	-2.110 **	-2.110 **	-2.120 **
AR(2)	0.110	-0.000	-0.000	-0.010	-0.010
No. instruments	41	41	41	41	41

Notes: Coef. denotes the coefficient of explanatory variables and Std. error represents the heteroscedasticity-robust asymptotic standard error. AR(1) is a serial correlation test of order 1 (and AR(2) of order 2 using residuals in first differences, under the null hypothesis of no serial correlation. Wald (χ^2) is a test of the joint significance of the reported coefficients, under the null hypothesis of no relationship. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

Regarding the control variables in the most robust regression model at the NUTS III regional, ht_location_church5, ht_events_meetings, f_efficiency_r, ht_room_ocup_nutsiii, and f_profit_r showed significant relationships with

the Z-score. The number of meeting and event rooms, operational efficiency, and room occupancy rates were positively associated with improved credit risk profiles. Conversely, total assets showed a negative association, suggesting that larger hotels show lower Z-Score, i.e. higher risk. Profitability and growth rates consistently contributed negatively to the Z-score, indicating their relevance for risk assessment. Market share concentration, liquidity, and debt ratios did not present consistent significance. The AR(1) test confirmed absence of first-order serial correlation, supporting model validity, and the Wald test indicated overall joint significance of the included variables.

5. CONCLUSIONS AND FUTURE RESEARCH

This thesis offers an in-depth investigation into the determinants of credit risk using Z-Score within the Portuguese hotel industry using a sample of MSMEs and BMEs in the period 2010–2015, emphasizing the importance of integrating both spatial and financial-operational factors to accurately assess a hotel's credit risk. The research highlights that location plays a significant role in influencing credit risk, with hotels situated closer to major tourist points such as beaches, castles, and churches generally exhibiting lower risk profiles. This proximity increases hotel's revenue by attracting more visitors, increasing occupancy rates, and providing a buffer against seasonal fluctuations, which collectively contribute to a hotel's financial stability. The study employs a dynamic panel data approach using the one-step System GMM estimator, with robustness confirmed through serial correlation tests, supporting the validity and reliability of the results. The Portuguese hotel sector experienced increased credit risk during the period influenced by the broader economic crisis affecting both Portugal and Europe. The sovereign debt crisis led to austerity measures, reduced domestic consumption, and restricted access to credit. These factors collectively weakened demand in the tourism and hospitality industries, increasing financial vulnerability for hotels. These variations highlight the need for context-specific risk assessments that consider both local and broader economic dynamics. Future research could extend this analysis by incorporating alternative credit risk measures such as the Altman-Z-Score with appropriate control variables, as well as by exploring other types of tourist points to better capture spatial effects. Additionally, future research could explore hotel-specific characteristics—such as the presence of golf facilities, swimming pools, star ratings, chain affiliation, and diversity of services—through interaction effects with location variables using the Z-Score could further clarify their impact on credit risk. Overall, this research demonstrates that an effective credit risk assessment in the hotel sector must go beyond traditional financial analysis. Incorporating spatial data on proximity to tourist attractions, operational performance indicators, and financial health metrics—such as total assets, efficiency, profit, and room occupancy—offers a more comprehensive and accurate evaluation of a hotel's risk profile. Efficiency exhibits a strong positive impact on the Z-score, indicating that hotels managing their resources effectively are less likely to face financial distress. Furthermore, room occupancy levels positively correlate with financial stability, highlighting the importance of operational performance. These findings suggest that lenders and investors should adopt an integrated approach that accounts for both geographic location and firm-specific characteristics when making informed decisions. Policymakers aiming to promote sustainable tourism development should also consider these factors to foster investment, enhance regional competitiveness, and support the stability of the hospitality industry. By recognizing the multifaceted nature of credit risk, stakeholders can better identify resilient hotels and implement targeted strategies to mitigate potential defaults, ultimately contributing to the long-term growth and sustainability of Portugal's tourism sector.

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ANNEXES

Annex 01 - Hotels' characteristics

We identified and analyzed detailed information on hotels registered in Portugal. [Table 10: Hotels' characteristics](#) presents the variables regarding characteristics of the hotels and definitions. The variables used from Turismo de Portugal dataset include: the hotel's name (`ht_name`), unique identifier for each hotel (`ht_id`), star classification (`ht_stars_classification`, `ht_quality_stars`), total capacity (`ht_capacity`), for which we computed the natural logarithm (`ht_ln_capacity`), the number of accommodation units (`ht_number_of_accommodation_units`), along with its logarithmic transformation (`ht_ln_accommodation`). The number of units adapted for reduced mobility (`ht_number_of_reduced_mobility_un`), we calculated its logarithm (`ht_ln_reducedmobility`), the ratio of reduced mobility units to total accommodation units (`ht_r_reducedmobility_totalaccomo`), and the ratio to total capacity (`ht_r_reducedmobility_totalcapaci`).

Location-related information included the specific address (`ht_address_location`) and postal code (`ht_postal_code`). The dataset also provided the municipality (`ht_municipality`), county (`ht_postal_designation`), and district (`ht_district`). Contact information included telephone (`ht_telephone_contact`), fax (`ht_fax_contact`), email (`ht_email_contact`), and website link (`ht_website`). Geographically, it contained the NUTS II region (`ht_nuts_2_newcriteria`), and the NUTS III region (`ht_nuts_3_newcriteria`). The dataset also included information on closure periods (`ht_closure_periods`), indicating whether a hotel had temporarily closed, and on touristic utility status (`ht_tourist_utility`), which identifies whether the hotel is classified for tourism purposes. Additional variables included the number of restaurants (`ht_number_of_restaurants`), for which we computed the logarithm (`ht_ln_restaurants`), and the availability of amenities such as meeting rooms (`ht_meeting_rooms`), their capacity (`ht_meeting_room_capacity`, `ht_events_meetings`), SPAs (`ht_spa`), outdoor and indoor pools (`ht_outdoor_pools`, `ht_indoor_pools`), tennis courts (`ht_tennis`), gyms (`ht_gym`), golf facilities (`ht_golf`), and other facilities (`ht_other_facilities`). Ownership and registration information included the managing entity name (`ht_managing_entity`), its identifier (`ht_managing_entity_id`), the year of opening (`ht_opening_year`), the Tax Identification Number (`ht_managing_entity_tax_id`, `f_tax_identification_number_tin`, `f_fiscal_number`), and the official registration number on the Turismo de Portugal website (`ht_registration_number`). Lastly, the dataset contained the type of tourism establishment (`ht_accommodation_type`), the hotel's brand (`ht_brands`), classification status (`ht_classification_status`) assigned by the responsible entity, the regional tourism authority (`ht_ert`), and the heritage classification level (`ht_classified_heritage_building`), indicating the type of interest attributed to the hotel—whether public, regional, national, local, or historical.

Then, the variable market share by municipality (`f_mkt_share_munic`) was created by calculating each hotel's share of total annual revenue within its respective municipality (`f_tot_oper_rev_turn_munic`). Total revenues were aggregated by municipality and year. Secondly, market concentration (`f_mkt_share_conc_munic`) was computed as the sum of squared market shares (`f_mkt_share_pwr_munic`), and the resulting values were assigned to each hotel based on its corresponding municipality and reporting year. Thirdly, the occupancy control variable (`ht_room_ocup_municv`) was developed using monthly room occupancy rates (`ht_room_ocup_rt_munic_1` to `_12`), averaged annually by municipality. Finally, seasonality (`ht_room_ocup_sea_munic`) was measured as the

variance of monthly occupancy rates, also aggregated by municipality and year. All variables were matched to each hotel according to its location and reporting year.

Lastly, we integrated data from Pordata on the number of overnight stays in Portugal from 2010 to 2015. Hotel records were indexed by year and geographic level—NUTS II, NUTS III, and municipality—and matched with the corresponding regional data from Pordata. The resulting variables were named (pdt_overnight_stays_nutsii_2024, pdt_overnight_stays_nutsiii_2024, and pdt_overnight_stays_muncic).

Table 10: Hotels' characteristics

Stata Variables	Dataset and Variables	Definition	Source
ht_name	Hotels name	Hotels Name.	Turismo de Portugal
ht_id	Hotels ID number	Hotels identifier number.	Turismo de Portugal
ht_stars_classification	Hotels stars classification	Hotels stars classification (1, 2, 3, 4, 5).	Turismo de Portugal
ht_quality_stars	Hotels quality stars	Hotels stars classification (1, 2, 3, 4, 5).	Turismo de Portugal
ht_capacity	Hotels capacity	Hotels maximum number of rooms.	Turismo de Portugal
ht_ln_capacity	Hotels natural logarithm of capacity	Hotels natural logarithm maximum number of rooms.	Turismo de Portugal
ht_number_of_accommodation_units	Hotels number of accommodations	Hotels number of accommodation units.	Turismo de Portugal
ht_ln_accommodation	Hotels natural logarithm number of accommodations	Hotels natural logarithm number of accommodation units.	Turismo de Portugal
ht_number_of_reduced_mobility_units	Hotels number of reduced mobility units	Hotels number of accommodation units, adapted for guests with reduced mobility.	Turismo de Portugal
ht_ln_reducedmobility	Hotels natural logarithm number of reduced mobility units	Hotels natural logarithm number of accommodation units, adapted for guests with reduced mobility.	Turismo de Portugal
ht_r_reducedmobility_totalaccomo	Hotels reduced mobility to total accommodation	Hotels ratio of reduced mobility units to total accommodation units.	Turismo de Portugal
ht_r_reducedmobility_totalcapaci	Hotels reduced mobility to total capacity	Hotels ratio of reduced mobility units to total capacity.	Turismo de Portugal
ht_address_location	Hotels address location	Hotels physical address.	Turismo de Portugal
ht_postal_code	Hotels postal code	Hotels postal code.	Turismo de Portugal
ht_municipality	Hotels municipality	Hotels municipality location.	Turismo de Portugal

ht_postal_designation	Hotels postal designation	Hotels postal designation location.	Turismo de Portugal
ht_district	Hotels district	Hotels district location.	Turismo de Portugal
ht_telephone_contact	Hotels telephone contact	Hotels telephone contact.	Turismo de Portugal
ht_fax_contact	Hotels fax contact	Hotels fax contact.	Turismo de Portugal
ht_email_contact	Hotels email contact	Hotels e-mail contact.	Turismo de Portugal
ht_website	Hotels website	Hotels website page.	Turismo de Portugal
ht_nuts_2_newcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos II 2024	Hotels Nomenclature of Territorial Units for Statistical Purposes II refers to a classification system used by the European Union to divide regions for statistical analysis and reporting purposes (Grande Lisboa, Algarve, Norte, Oeste e Vale do Tejo, Centro, Região Autónoma da Madeira, Alentejo, Região Autónoma dos Açores, Península de Setúbal.).	Turismo de Portugal
ht_nuts_3_newcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos II 2024	Hotels Portugal Nomenclature of Territorial Units for Statistical Purposes II per categories numbers (Alto Minho, Cávado, Ave, Área Metropolitana do Porto, Alto Tâmega, Tâmega e Sousa, Douro, Terras de Trás-os-Montes, Algarve, Região de Aveiro, Região de Coimbra, Região de Leiria, Viseu Dão-Lafões, Beira Baixa, Beiras e Serra da Estrela, Grande Lisboa, Península de Setúbal, Alentejo Litoral, Baixo Alentejo, Alto Alentejo, Alentejo Central, Oeste, Médio Tejo, Lezíria do Tejo, Região Autónoma dos Açores, Região Autónoma da Madeira.).	Turismo de Portugal
ht_closure_periods	Hotels closure periods	Hotels closure periods.	Turismo de Portugal
ht_tourist_utility	Hotels tourist utility	Hotels tourist utility the official authorization or certified completion allowing the establishment to begin operating for tourism purposes (Yes, and No).	Turismo de Portugal
ht_number_of_restaurants	Hotels number of restaurants	Hotels number of restaurants.	Turismo de Portugal
ht_ln_restaurants	Hotels natural logarithm number of restaurants	Hotels natural logarithm number of restaurants.	Turismo de Portugal
ht_meeting_rooms	Hotels meeting rooms	Hotels meeting rooms availability (Yes, No).	Turismo de Portugal
ht_meeting_rooms_capacity	Hotels meeting rooms capacity	Hotels number of meeting rooms capacity.	Turismo de Portugal

ht_events_meetings	Hotels events meeting	Hotels number of meeting rooms capacity.	Turismo de Portugal
ht_spas	Hotels spa	Hotels SPA availability (Yes, No).	Turismo de Portugal
ht_outdoor_pools	Hotels outdoor pools	Hotels Outdoor Pools availability (Yes, No).	Turismo de Portugal
ht_indoor_pools	Hotels indoor pools	Hotels Indoor Pools availability (Yes, No).	Turismo de Portugal
ht_tennis	Hotels tennis	Hotels Tennis availability (Yes, No).	Turismo de Portugal
ht_gym	Hotels gym	Hotels Gym availability (Yes, No).	Turismo de Portugal
ht_golf	Hotels golf	Hotels Golf availability (Yes, No).	Turismo de Portugal
ht_other_facilities	Hotels other facilities	Hotels Other Facilities availability (Yes, No).	Turismo de Portugal
ht_managing_entity	Hotels managing entities	Hotels exploring entity name.	Turismo de Portugal
ht_managing_entity_id	Hotel managing entities ID	Hotel exploring managing entities identifier number.	Turismo de Portugal
ht_opening_year	Hotels opening year	Hotels opening year.	Turismo de Portugal
ht_managing_entity_tax_id	Hotel managing entity tax number	Entity tax identification number.	Turismo de Portugal
f_tax_identification_number_tin	Hotels Tax Identification Number	Entity tax identification number.	Orbis
ht_registration_number	Hotels registration number	Hotels serial number assigned within National Touristic Enterprise National Register.	Turismo de Portugal
ht_accommodation_type	Hotels accommodation type	Classification of the tourism enterprise based on its accommodation typology (Hotels Establishment - Hotels).	Turismo de Portugal
ht_brands	Hotels brands	Hotels brand or affiliated chain.	Turismo de Portugal
ht_classification_status	Hotels classification status	Hotels stars classification their status (Classificado em auditoria, Classificado em projeto, Por reconverter).	Turismo de Portugal
ht_ert	Hotels Entidade Regionais de Turismo	Hotels tourism regional entity (Entidade Regional de Turismo da Região de Lisboa, Região de Turismo do Algarve, Secretaria Regional da Energia, Ambiente e Turismo dos Açores, Secretaria Regional de Economia, Turismo e Cultura da Madeira, Turismo do Alentejo e do Ribatejo, E.R.T., Turismo do Centro de Portugal, Turismo do Porto e Norte de Portugal).	Turismo de Portugal
ht_classified_heritage_building	Hotels classified heritage	Hotels entity type interest classification (---; from epoch; Local; National; Public; Regional)	Turismo de Portugal
ht_room_occupancy_rate_nutsii_1	Hotels room occupation rate NUTS II January	Hotels' room occupancy rate within each NUTS II region in January.	Turismo de Portugal

ht_room_ocup_rt_nutsii_2	Hotels room occupation rate NUTS II February	Hotels' room occupancy rate within each NUTS II region in February.	Turismo de Portugal
ht_room_ocup_rt_nutsii_3	Hotels room occupation rate NUTS II March	Hotels' room occupancy rate within each NUTS II region in March.	Turismo de Portugal
ht_room_ocup_rt_nutsii_4	Hotels room occupation rate NUTS II April	Hotels' room occupancy rate within each NUTS II region in April.	Turismo de Portugal
ht_room_ocup_rt_nutsii_5	Hotels room occupation rate NUTS II May	Hotels' room occupancy rate within each NUTS II region in May.	Turismo de Portugal
ht_room_ocup_rt_nutsii_6	Hotels room occupation rate NUTS II June	Hotels' room occupancy rate within each NUTS II region in June.	Turismo de Portugal
ht_room_ocup_rt_nutsii_7	Hotels room occupation rate NUTS II July	Hotels' room occupancy rate within each NUTS II region in July.	Turismo de Portugal
ht_room_ocup_rt_nutsii_8	Hotels room occupation rate NUTS II August	Hotels' room occupancy rate within each NUTS II region in August.	Turismo de Portugal
ht_room_ocup_rt_nutsii_9	Hotels room occupation rate NUTS II September	Hotels' room occupancy rate within each NUTS II region in September.	Turismo de Portugal
ht_room_ocup_rt_nutsii_10	Hotels room occupation rate NUTS II October	Hotels' room occupancy rate within each NUTS II region in October.	Turismo de Portugal
ht_room_ocup_rt_nutsii_11	Hotels room occupation rate NUTS II November	Hotels' room occupancy rate within each NUTS II region in November.	Turismo de Portugal
ht_room_ocup_rt_nutsii_12	Hotels room occupation rate NUTS II December	Hotels' room occupancy rate within each NUTS II region in December.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_1	Hotels room occupation rate NUTS III January	Hotels' room occupancy rate within each NUTS III region in January.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_2	Hotels room occupation rate NUTS III February	Hotels' room occupancy rate within each NUTS III region in February.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_3	Hotels room occupation rate NUTS III March	Hotels' room occupancy rate within each NUTS III region in March.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_4	Hotels room occupation rate NUTS III April	Hotels' room occupancy rate within each NUTS III region in April.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_5	Hotels room occupation rate NUTS III May	Hotels' room occupancy rate within each NUTS III region in May.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_6	Hotels room occupation rate NUTS III June	Hotels' room occupancy rate within each NUTS III region in June.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_7	Hotels room occupation rate NUTS III July	Hotels' room occupancy rate within each NUTS III region in July.	Turismo de Portugal

ht_room_ocup_rt_nutsiii_8	Hotels room occupation rate NUTS III August	Hotels' room occupancy rate within each NUTS III region in August.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_9	Hotels room occupation rate NUTS III September	Hotels' room occupancy rate within each NUTS III region in September.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_10	Hotels room occupation rate NUTS III October	Hotels' room occupancy rate within each NUTS III region in October.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_11	Hotels room occupation rate NUTS III November	Hotels' room occupancy rate within each NUTS III region in November.	Turismo de Portugal
ht_room_ocup_rt_nutsiii_12	Hotels room occupation rate NUTS III December	Hotels' room occupancy rate within each NUTS III region in December.	Turismo de Portugal
ht_room_ocup_rt_munic_1	Hotels room occupation rate municipalities January	Hotels' room occupancy rate within each municipality region in January.	Turismo de Portugal
ht_room_ocup_rt_munic_2	Hotels room occupation rate municipalities February	Hotels' room occupancy rate within each municipality region in February.	Turismo de Portugal
ht_room_ocup_rt_munic_3	Hotels room occupation rate municipalities March	Hotels' room occupancy rate within each municipality region in March.	Turismo de Portugal
ht_room_ocup_rt_munic_4	Hotels room occupation rate municipalities April	Hotels' room occupancy rate within each municipality region in April.	Turismo de Portugal
ht_room_ocup_rt_munic_5	Hotels room occupation rate municipalities May	Hotels' room occupancy rate within each municipality region in May.	Turismo de Portugal
ht_room_ocup_rt_munic_6	Hotels room occupation rate municipalities June	Hotels' room occupancy rate within each municipality region in June.	Turismo de Portugal
ht_room_ocup_rt_munic_7	Hotels room occupation rate municipalities July	Hotels' room occupancy rate within each municipality region in July.	Turismo de Portugal
ht_room_ocup_rt_munic_8	Hotels room occupation rate municipalities August	Hotels' room occupancy rate within each municipality region in August.	Turismo de Portugal
ht_room_ocup_rt_munic_9	Hotels room occupation rate municipalities September	Hotels' room occupancy rate within each municipality region in September.	Turismo de Portugal
ht_room_ocup_rt_munic_10	Hotels room occupation rate municipalities October	Hotels' room occupancy rate within each municipality region in October.	Turismo de Portugal
ht_room_ocup_rt_munic_11	Hotels room occupation rate municipalities November	Hotels' room occupancy rate within each municipality region in November.	Turismo de Portugal
ht_room_ocup_rt_munic_12	Hotels room occupation rate municipalities December	Hotels' room occupancy rate within each municipality region in December.	Turismo de Portugal

ht_room_ocup_nutsii	Hotel room occupancy rate average NUTS II	Hotels' average monthly room occupancy rate within each NUTS II region.	Turismo de Portugal
ht_room_ocup_nutsiii	Hotel room occupancy rate average NUTS III	Hotels' average monthly room occupancy rate within each NUTS III region.	Turismo de Portugal
ht_room_ocup_municv	Hotel room occupancy rate average municipalities	Hotels' average monthly room occupancy rate within each municipality region.	Turismo de Portugal
ht_bed_ocup_rt_nutsii_y	Hotel bedroom occupancy rate (annual, NUTS II)	Hotels' annual bedroom occupancy rate by NUTS II region.	Turismo de Portugal
ht_occupancy_beds_nuts_3_newcrit	Hotel bedroom occupancy rate (annual, NUTS III)	Hotels' annual bedroom occupancy rate by NUTS III region.	Turismo de Portugal
ht_room_ocup_sea_nutsii	Hotel room occupancy rate variance NUTS II	Hotels' variance monthly room occupancy rate within each NUTS II region.	Turismo de Portugal
ht_room_ocup_sea_nutsiii	Hotel room occupancy rate variance NUTS III	Hotels' variance monthly room occupancy rate within each NUTS III region.	Turismo de Portugal
ht_room_ocup_sea_munic	Hotel room occupancy rate variance municipalities	Hotels' variance monthly room occupancy rate within each municipal region.	Turismo de Portugal
pdt_overnight_stays_nutsii_2024	Hotel number of overnight stays NUTS II	Hotel's number of overnight stays by NUTS II region.	Pordata
pdt_overnight_stays_nutsiii_2024	Hotel number of overnight stays NUTS III	Hotel's number of overnight stays by NUTS III region.	Pordata
pdt_overnight_stays_munic	Hotel number of overnight stays municipalities	Hotel's number of overnight stays by municipality region.	Pordata
ht_location_1	Hotels location 1	Hotel minimum distance from all touristic points.	Visit Portugal
ht_location_2	Hotels location 2	Hotels average 3 minimum distances (first, second, and third Euclidean minimum distances) from all touristic points (castles, beaches, churches).	Visit Portugal
ht_location_3	Hotels location 3	Hotels average of the 6 minimum distances (first, second, third, fourth, fifth, sixth Euclidean minimum distances) from all touristic points (castles, beaches, churches).	Visit Portugal
ht_location_4	Hotels location 4	Hotels average 9 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth Euclidean minimum distances) from all touristic points (castles, beaches, churches).	Visit Portugal
ht_location_5	Hotels location 5	Hotels average of the 12 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth Euclidean minimum distances) from all touristic points (castles, beaches, churches).	Visit Portugal

ht_location_beach 1	Hotels location beach 1	Hotel minimum distance from all beaches points.	Visit Portugal
ht_location_beach 2	Hotels location beach 2	Hotels average the 3 minimum distances (first, second, and third Euclidean minimum distances) from all beaches points.	Visit Portugal
ht_location_beach 3	Hotels location beach 3	Hotels average of 6 minimum distances (first, second, third, fourth, fifth, sixth Euclidean minimum distances) from all beaches points.	Visit Portugal
ht_location_beach 4	Hotels location beach 4	Hotels average the 9 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth Euclidean minimum distances) from all beaches points.	Visit Portugal
ht_location_beach 5	Hotels location beach 5	Hotels average of the 12 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth Euclidean minimum distances) from all beaches points.	Visit Portugal
ht_location_castle 1	Hotels location castle 1	Hotel minimum distance from all castles points.	Visit Portugal
ht_location_castle 2	Hotels location castle 2	Hotels average of the 3 minimum distances (first, second, and third Euclidean minimum distances) from all castles points.	Visit Portugal
ht_location_castle 3	Hotels location castle 3	Hotels average of the 6 minimum distances (first, second, third, fourth, fifth, sixth Euclidean minimum distances) from all castles points.	Visit Portugal
ht_location_castle 4	Hotels location castle 4	Hotels average of the 9 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth Euclidean minimum distances) from all castles points.	Visit Portugal
ht_location_castle 5	Hotels location castle 5	Hotels average of 12 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth Euclidean minimum distances) from all castles points.	Visit Portugal
ht_location_church h1	Hotels location church 1	Hotel minimum distance from all churches points.	Visit Portugal
ht_location_church h2	Hotels location church 2	Hotels average 3 minimum distances (first, second, and third Euclidean minimum distances) from all churches points.	Visit Portugal
ht_location_church h3	Hotels location church 3	Hotels average of 6 minimum distances (first, second, third, fourth, fifth, sixth Euclidean minimum distances) from all churches points.	Visit Portugal

ht_location_church4	Hotels location church 4	Hotels average of the 9 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth Euclidean minimum distances) from all churches points.	Visit Portugal
ht_location_church5	Hotels location church 5	Hotels average of 12 minimum distances (first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth Euclidean minimum distances) from all churches points.	Visit Portugal
ht_secondmineuclidean	Hotels second minimum Euclidean distance	Hotels second minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_thirdmineuclidean	Hotels third minimum Euclidean distance	Hotels third minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_fourthmineuclidean	Hotels fourth minimum Euclidean distance	Hotels fourth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_fifthmineuclidean	Hotels fifth minimum Euclidean distance	Hotels fifth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_sixthmineuclidean	Hotels sixth minimum Euclidean distance	Hotels sixth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_seventhmineuclidean	Hotels seventh minimum Euclidean distance	Hotels seventh minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_eighthmineuclidean	Hotels eighth minimum Euclidean distance	Hotels eighth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_ninthmineuclidean	Hotels ninth minimum Euclidean distance	Hotels ninth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_tenthmineuclidean	Hotels tenth minimum Euclidean distance	Hotels tenth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_eleventhmineuclidean	Hotels eleventh minimum Euclidean distance	Hotels eleventh minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_twelfthmineuclidean	Hotels twelfth minimum Euclidean distance	Hotels twelfth minimum distance from all touristic points (castles, beaches, churches).	Visit Portugal
ht_secondmineuclidean_bc	Hotels second minimum Euclidean distance beaches	Hotel second minimum distance from all beaches points.	Visit Portugal
ht_ThirdMinEuclideanDist_bch	Hotels third minimum Euclidean distance beaches	Hotel third minimum distance from all beaches points.	Visit Portugal
ht_FourthMinEuclideanDist_bch	Hotels fourth minimum Euclidean distance beaches	Hotel fourth minimum distance from all beaches points.	Visit Portugal
ht_FifthMinEuclideanDist_bch	Hotels fifth minimum Euclidean distance beaches	Hotel fifth minimum distance from all beaches points.	Visit Portugal

ht_SixthMinEuclidDist_bch	Hotels sixth minimum Euclidean distance beaches	Hotel sixth minimum distance from all beaches points.	Visit Portugal
ht_seventhmineuclidanddistance_b	Hotels seventh minimum Euclidean distance beaches	Hotel seventh minimum distance from all beaches points.	Visit Portugal
ht_eighthmineuclidanddistance_bc	Hotels eighth minimum Euclidean distance beaches	Hotel eighth minimum distance from all beaches points.	Visit Portugal
ht_ninthmineuclidanddistance_bc	Hotels ninth minimum Euclidean distance beaches	Hotel ninth minimum distance from all beaches points.	Visit Portugal
ht_tenthmineuclidanddistance_bch	Hotels tenth minimum Euclidean distance beaches	Hotel tenth minimum distance from all beaches points.	Visit Portugal
ht_eleventhmineuclidanddistance_	Hotels eleventh minimum Euclidean distance beaches	Hotel eleventh minimum distance from all beaches points.	Visit Portugal
ht_twelfthmineuclidanddistance_b	Hotels twelfth minimum Euclidean distance beaches	Hotel twelfth minimum distance from all beaches points.	Visit Portugal
ht_SecondMinEuclidDist_ctl	Hotels second minimum Euclidean distance castles	Hotel second minimum distance from all castles points.	Visit Portugal
ht_ThirdMinEuclidDist_ctl	Hotels third minimum Euclidean distance castles	Hotel third minimum distance from all castles points.	Visit Portugal
ht_FourthMinEuclidDist_ctl	Hotels fourth minimum Euclidean distance castles	Hotel fourth minimum distance from all castles points.	Visit Portugal
ht_FifthMinEuclidDist_ctl	Hotels fifth minimum Euclidean distance castles	Hotel fifth minimum distance from all castles points.	Visit Portugal
ht_SixthMinEuclidDist_ctl	Hotels sixth minimum Euclidean distance castles	Hotel sixth minimum distance from all castles points.	Visit Portugal
ht_seventhmineuclidanddistance_c	Hotels seventh minimum Euclidean distance castles	Hotel seventh minimum distance from all castles points.	Visit Portugal
ht_eighthmineuclidanddistance_ct	Hotels eighth minimum Euclidean distance castles	Hotel eighth minimum distance from all castles points.	Visit Portugal
ht_ninthmineuclidanddistance_ct	Hotels ninth minimum Euclidean distance castles	Hotel ninth minimum distance from all castles points.	Visit Portugal
ht_tenthmineuclidanddistance_ctl	Hotels tenth minimum Euclidean distance castles	Hotel tenth minimum distance from all castles points.	Visit Portugal
ht_EleventhMinEuclidDist_ctlv	Hotels eleventh minimum Euclidean distance castles	Hotel eleventh minimum distance from all castles points.	Visit Portugal
ht_twelfthmineuclidanddistance_c	Hotels twelfth minimum Euclidean distance castles	Hotel twelfth minimum distance from all castles points.	Visit Portugal

ht_SecondthMinEuclidDist_chc	Hotels second minimum Euclidean distance churches	Hotel second minimum distance from all churches points.	Visit Portugal
ht_ThirdMinEuclidDist_chc	Hotels third minimum Euclidean distance churches	Hotel third minimum distance from all churches points.	Visit Portugal
ht_FourthMinEuclidDist_chc	Hotels fourth minimum Euclidean distance churches	Hotel fourth minimum distance from all churches points.	Visit Portugal
ht_FifthMinEuclidDist_chc	Hotels fifth minimum Euclidean distance churches	Hotel fifth minimum distance from all churches points.	Visit Portugal
ht_SixthMinEuclidDist_chc	Hotels sixth minimum Euclidean distance churches	Hotel sixth minimum distance from all churches points.	Visit Portugal
ht_SeventhMinEuclidDist_chc	Hotels seventh minimum Euclidean distance churches	Hotel seventh minimum distance from all churches points.	Visit Portugal
ht_eighthmineuclidanddistance_ch	Hotels eighth minimum Euclidean distance churches	Hotel eighth minimum distance from all churches points.	Visit Portugal
ht_ninthmineuclidanddistance_ch	Hotels ninth minimum Euclidean distance churches	Hotel ninth minimum distance from all churches points.	Visit Portugal
ht_tenthmineuclidanddistance_chc	Hotels tenth minimum Euclidean distance churches	Hotel tenth minimum distance from all churches points.	Visit Portugal
ht_ElevMinEuclideanDistance_chc	Hotels eleventh minimum Euclidean distance churches	Hotel eleventh minimum distance from all churches points.	Visit Portugal
ht_twelfththmineuclidanddistance	Hotels twelfth minimum Euclidean distance churches	Hotel twelfth minimum distance from all churches points.	Visit Portugal

Annex 02 - Hotels' Financial variables

Financial indicators include net income before deduction of minority interests ($f_p_1_for_period_net_income$) and its Winsorized at significance level 5% version ($f_profit_loss_for_the_period_n0$). We also included total assets (f_total_assets , $f_total_assets_w$), which represent hotels' owned resources (fixed assets plus current assets), their natural logarithm ($f_ln_total_assets$, $f_ln_total_assets_w$), and issued share capital ($f_capital$, $f_capital_w$), which represents total funds raised by the company through issued share capital. Asset composition variables comprise

non-current assets (*f_noncurrentassets*), which are long-term assets not expected to be converted to cash within one year; intangible assets (*f_intangibleassets*), which include formation expenses, research expenses, goodwill, development expenses, and other long-term effects; tangible fixed assets (*f_tangiblefixedassets*), referring to long-term physical assets acquired to support the company's operations; and current assets (*f_currentassets*), which are short-term assets convertible to cash within a year (stocks, debtors, other assets). Additional asset variables include stock (*f_stock*), representing total inventory including raw materials, work in progress, and finished goods, and debtors (*f_debtors*), which refer to receivables from clients and customers for services provided. Return on assets is captured through several variables (*f_roa*, *f_roa_w*, *f_roa_v1*, *f_roa_v2*). Other current assets (*f_othercurrentassets*) encompass receivables such as taxes and intra-group balances, short-term investments, and cash at bank and on hand. Cash and cash equivalents (*f_ofwhichcashandcashequivalent*) represent the amount of cash at bank and in hand. Equity-related indicators include total shareholders' funds (*f_shareholders_funds*), reflecting total equity including capital and other shareholder contributions, and other shareholders' funds (*f_other_shareholders_funds*), which represent equity components not linked to issued capital, such as reserves, retained earnings, and minority interests. Revenue-related data include operating revenue excluding Value Added Tax Number (*f_operating_revenue_turnover*, *f_operating_revenue_turnover2*), which represent total operating revenues calculated as the sum of net sales, other operating revenues, and stock variations excluding VAT. Net sales (*f_sales*) correspond to revenue from goods and services sold, after deducting returns, discounts, and allowances.

Liabilities are divided into non-current liabilities (*f_non_current_liabilities*), referring to long-term liabilities such as financial debts, other long-term obligations, and provisions. These include long-term debt (*f_long_term_debt*), which covers loans, credits, and bonds; other long-term obligations; and short-term debt (*f_loans_short_term_debt*), which are borrowed funds and obligations due within one year. Other non-current liabilities (*f_other_non_current_liabilities*) include trade debts, group company obligations, pension loans, additional provisions, and deferred taxes. Current liabilities (*f_current_liabilities*) consist of short-term obligations including loans, creditors, and other current liabilities. These are further broken down into trade creditors (*f_creditors*), representing debts to suppliers and contractors, and other short-term liabilities (*f_other_current_liabilities*), which include pensions, personnel costs, taxes, intra-group debts, and advance payments. Total shareholders' funds and liabilities (*f_ttl_shareholders_funds_and_lia*) represent the sum of shareholders' funds, non-current liabilities, and current liabilities. Cost components include the cost of goods sold (*f_costs_of_goods_sold*), which are direct costs of goods, production, and services including related depreciation, and other operating expenses (*f_other_operating_expense_income*), which represent costs not directly related to production, such as commercial and administrative expenses plus depreciation. The debt ratio (*f_debt_r*), which represents total liabilities to shareholders equity ratio.

We also captured gross profit (*f_gross_profit*), calculated as operating revenue minus cost of goods sold; earnings before interest and taxes (*f_ebit*), which is all operating revenues minus all operating expenses; financial profit/loss (*f_financial_profit_loss*), which includes financial revenue minus financial expenses; financial revenue (*f_financial_revenue*), covering interest, income from shares, and other financial earnings; financial expenses (*f_financial_expenses*), including interest charges and write-offs of financial assets; profit before tax

(f_profit_loss_before_tax_pbt, f_p_l_before_tax), representing profit from operating profit plus financial profit; and profit after tax (f_profit_l_after_tax_pat), calculated as profit before taxation minus taxation.

Liquidity and solvency indicators include the current ratio (f_current_ratio) and the asset-based solvency ratio (f_solvency_ratio_asset_based). The efficiency ratio (f_efficiency_r) is calculated as the ratio of income to total assets. The growth ratio (f_growth_r) represents the ratio of hotel revenues in the current period to those in the previous period. Lastly, the liquidity ratio (f_liquidity_r) is defined as the ratio of current assets (excluding stocks) to current liabilities. Financial ratios include return on equity (f_roe, f_roe_w, f_roe_v1, f_roe_v2), based on profit loss before tax (f_roe_using_pft_loss_before_tax, f_roe_using_p_l_before_tax) and net income (f_roe_using_net_income). Return on capital employed is measured using profit loss before tax (f_roce_using_pft_loss_before_tax, f_roce_using_p_l_before_tax) and net income (f_roce_using_net_income). Return on assets is calculated both on profit loss before tax (f_roa_using_pft_loss_before_tax) and net income (f_roa_using_net_income). Various margin metrics include profit margin (f_profit_margin), gross margin (f_gross_margin), earnings before interest, taxes, depreciation, and amortization margin (f_ebitda_margin), and earnings before interest and taxes margin (f_ebit_margin, f_profit_r). Cash flow operating revenue (f_cash_flow_operating_revenue) captures net income and depreciation divided by operating revenues.

Table 11: Hotels' Financial variables

Stata Variables	Dataset and Variables	Definition	Source
f_fiscal_number	Hotel fiscal number	Entity tax identification number.	Orbis
f_p_l_for_period_net_income	Hotels profit loss for the period = (Net Income)	Hotels Net Income for the year. Before deduction of minority interests if any (Profit after taxation + Extraordinary and other profit).	Orbis
f_total_assets	Hotels total assets	Hotels total owned resources (Fixed Assets + Current Assets).	Orbis
f_total_assets_w	Hotels total assets Winsorized at significance level 5%	Hotels Winsorized at significance level 5% total owned resources (Fixed Assets + Current Assets).	Orbis
f_ln_total_assets	Hotels natural logarithm of total assets	Hotels logarithm total owned resources (Fixed Assets + Current Assets).	Orbis
f_ln_total_assets_w	Hotels natural logarithm of total assets Winsorized at significance level 5%	Hotels Winsorized at significance level 5% logarithm total owned resources (Fixed Assets + Current Assets).	Orbis
f_capital	Hotels capital	Hotels total money raised by the company through issued shares capital (Authorized capital).	Orbis
f_capital_w	Hotels capital Winsorized at significance level 5%	Hotels total money raised by the company through issued shares capital (Authorized capital) Winsorized at significance level 5%.	Orbis
f_noncurrent_assets	Hotels non-current assets	Hotels long-term assets that are not expected to be converted to cash within one year.	Orbis
f_intangible_assets	Hotels intangible assets	Hotels all intangible assets such as formation expenses, research expenses, goodwill, development expenses and all other expenses with a long term effect.	Orbis

f_tangiblefixedassets	Hotels tangible fixed assets	Hotels long-term physical assets acquired to support a company's operations such as buildings, machinery.	Orbis
f_currentassets	Hotels current assets	Hotels short-term assets convertible to cash within a year (stocks, debtors, other assets).	Orbis
f_stock	Hotels stocks	Hotels total inventory, including raw materials, work in progress, and finished goods.	Orbis
f_debtors	Hotels debtors	Hotels receivables from clients and customers for services provided.	Orbis
f_roa	Hotels return on assets	Hotels return on assets (Net Income/ Total Assets) x 100	Orbis
f_roa_w	Hotels return on assets Winsorized at significance level 5%	Hotels return on assets Winsorized at significance level 5% (Net Income Winsorized at significance level 5%/ Total Assets Winsorized at significance level 5%) x 100	Orbis
f_roa_v1	Hotels return on assets ratio 1	Hotels return on assets ratio (Return on Assets using Profit Loss before Tax / 100)	Orbis
f_roa_v2	Hotels return on assets ratio 2	Hotels return on assets ratio (Return on Assets using Net Income/ 100)	Orbis
f_othercurrentassets	Hotels other current assets	Hotels all other current assets, including receivables (e.g., taxes, group companies), short-term investments, and cash at bank and on hand.	Orbis
f_ofwhichcashandcashequivalent	Hotels of which cash and cash equivalent	Hotels amount of cash at bank and in hand.	Orbis
f_shareholders_funds	Hotels shareholders funds	Hotels total equity, including capital and other shareholder funds.	Orbis
f_other_shareholders_funds	Hotels other shareholders funds	Hotels shareholders funds not related to issued capital, including reserves, undistributed profits, and minority interests.	Orbis
f_operating_revenue_turnover2	Hotels operating revenue (turnover) 2	Hotels total operating revenues calculated as the sum of (net sales, other operating revenues, and stock variations) excluding Value-Added Tax).	Orbis
f_sales	Hotels sales	Hotels revenue from goods and services sold, after deducting returns, discounts, and allowances.	Orbis
f_non_current_liabilities	Hotels non-current liabilities	Hotels long-term liabilities, including financial debts, other long-term obligations, and provisions.	Orbis
f_long_term_debt	Hotels long term debt	Hotels long-term financial obligations, including loans, credits, and bonds.	Orbis
f_loans_short_term_debt	Hotels loans short term debt	Hotel borrowed funds and obligations due for repayment within one year.	Orbis
f_other_non_current_liabilities	Hotels other non-current liabilities	Hotels other long-term liabilities, including (trade debts, group company obligations, and pension loans) more provisions, and deferred taxes.	Orbis
f_current_liabilities	Hotels current liabilities	Hotels short-term obligations, including loans, creditors, and other current liabilities.	Orbis
f_creditors	Hotels creditors	Hotels debts to suppliers and contractors (trade creditors).	Orbis
f_other_current_liabilities	Hotels other current liabilities	Hotels other current liabilities include pensions, personnel costs, taxes, intra-group debts, and advance payments.	Orbis

f_ttl_shareholders_funds_and_lia	Hotels total shareholders' funds and liabilities	Hotels total funds and liabilities, calculated as the sum of shareholders' funds, non-current liabilities, and current liabilities.	Orbis
f_costs_of_goods_sold	Hotels costs of goods sold	Hotels direct costs of goods, production, and services, including related depreciation of those costs.	Orbis
f_other_operating_expense_income	Hotels other operating expense income	Hotels costs are not directly related to the production of goods, such as commercial and administrative expenses, more the depreciation of these costs.	Orbis
f_gross_profit	Hotels gross profit	Hotels profit after deducting the costs associated with producing and selling its products or services calculated as (operating revenue minus cost of goods sold).	Orbis
f_ebit	Hotels earnings before interest and taxes	Hotels all operating revenues minus all operating expenses (gross profit minus other operating expenses).	Orbis
f_financial_profit_loss	Hotels financial profit loss	Hotels all financial activities (financial revenue minus financial expenses).	Orbis
f_financial_revenue	Hotels financial revenue	Hotels financial revenues include interest, income from shares, and other financial earnings.	Orbis
f_financial_expenses	Hotels financial expenses	Hotels financial expenses include interest charges and write-offs of financial assets.	Orbis
f_p_l_before_tax	Hotels profit loss before tax	Hotels profit that result from (operating profit plus financial profit).	Orbis
f_profit_l_after_tax_pat	Hotels profit loss after tax	Hotels profit after taxation that result from (profit before taxation - taxation).	Orbis
f_current_ratio	Hotels current ratio	Hotels Ratio (Current Assets / Current Liabilities)	Orbis
f_solvency_ratio_asset_based	Hotels solvency ratio asset based	Hotels Ratio (Shareholders funds / Total assets) x 100.	Orbis
f_roe	Hotels return on equity	Return on equity for hotels is the percentage of profit generated from shareholders investments, calculated as (Net Income / Shareholder Funds) × 100.	Orbis
f_roe_w	Hotels return on equity Winsorized at significance level 5%	Return on equity Winsorized at significance level 5% for hotels is the percentage of profit generated from shareholders investments, calculated as (Net Income / Shareholder Funds) × 100.	Orbis
f_roe_v1	Hotels return on equity ratio 1	Return on equity for hotels is the percentage of profit generated from shareholders investments, calculated as (Return on Equity using Profit Loss before Tax / 100).	Orbis
f_roe_v2	Hotels return on equity ratio 2	Return on equity for hotels is the percentage of profit generated from shareholders investments, calculated as (Return on Equity using Net Income/ 100).	Orbis
f_roe_using_pft_loss_before_tax	Hotels return on equity (using profit loss before tax)	Hotels Ratio (Profit loss before tax / Shareholders funds) x 100.	Orbis
f_roe_using_pft_loss_before_tax	Hotels return on equity (using profit loss before tax)	Hotels Ratio (Profit loss before tax / Shareholders funds) x 100.	Orbis
f_roe_using_net_income	Hotels return on equity (using Net Income)	Hotels Ratio (Net income / Shareholder funds) x 100.	Orbis

f_roce_using_pft_loss_before_tax	Hotels return on capital employed (using profit loss before tax)	Hotels Ratio (Profit loss before tax + Interest paid) / (Shareholders funds + Non-current liabilities) x 100.	Orbis
f_roce_using_p_l_before_tax	Hotels return on capital employed (using profit loss before tax) 2	Hotels Ratio (Profit loss before tax + Interest paid) / (Shareholders funds + Non-current liabilities) x 100.	Orbis
f_roce_using_net_income	Hotels return on capital employed (using Net Income)	Hotels Ratio (Net income + Interest paid) / (Shareholder funds + Non-current liabilities) x 100.	Orbis
f_roa_using_pft_loss_before_tax	Hotels return on assets (using profit loss before tax)	Hotels Ratio (Profit loss before tax / Total assets) x 100.	Orbis
f_roa_using_net_income	Hotels return on assets (using Net Income)	Hotels Ratio (Net income / Total Assets) x 100.	Orbis
f_profit_margin	Hotels profit margin	Hotels Ratio (Profit loss before tax / Operating revenue) x 100.	Orbis
f_gross_margin	Hotels gross margin	Hotels Ratio (Gross Profit / Operating revenue) x 100.	Orbis
f_ebitda_margin	Hotels earnings before interests taxes depreciation and amortization (EBITDA) margin	Hotels Ratio (EBITDA / Operating revenue) x 100.	Orbis
f_ebit_margin	Hotels earnings before interests and taxes (EBIT) margin	Hotels Ratio (EBIT / Operating revenue) x 100.	Orbis
f_cash_flow_operating_revenue	Hotels cash flow operating revenue	Hotels Ratio (Cash flow / Operating revenue) x 100.	Orbis
f_debt_r	Hotels debt ratio	Hotels ratio of total liabilities to shareholders equity.	Orbis
f_efficiency_r	Hotels efficiency ratio	Hotels ratio of income to total assets for each hotel.	Orbis
f_growth_r	Hotels growth ratio	Hotels ratio of revenue in period t to hotel revenue in period t-1.	Orbis
f_liquidity_r	Hotels liquidity ratio	Hotels ratio of current assets excluding stocks to current liabilities.	Orbis
f_profit_r	Hotels profit ratio	Hotels ratio of earnings before interest and taxes to revenue.	Orbis
f_mkt_share_conc_munic	Hotels market share concentration municipality	Sum of squared market shares of all hotels within each municipality.	Orbis
f_zscore_w	Hotels Z-score Winsorized at significance level 5%	Hotels Z-Score Winsorized at significance level 5% identifies which hotels with elevated risks of financial distress - (Return on Average Assets Winsorized at significance level 5% + Capital Ratio Winsorized at significance level 5%) / (Return on Average Assets Standard Deviation Winsorized at significance level 5%).	Orbis
f_mkt_share_conc_nutsii	Hotels market share concentration NUTS II	Sum of squared market shares of all hotels within each NUTS II region.	Orbis
f_roaa	Hotels return on average assets	Hotels Return on Average Assets (Net Income / (Total Assets (current year) + Total Assets (previous year))/2).	Orbis
f_mkt_share_conc_nutsiii	Hotels market share concentration NUTS III	Sum of squared market shares of all hotels within each NUTS III region.	Orbis

f_mkt_share_munic	Hotels market share municipality	Hotels ratio of each hotel's revenue to the total hotel revenues within the corresponding municipality.	Orbis
f_mkt_share_nutsii	Hotels market share NUTS II	Hotels ratio of each hotel's revenue to the total hotel revenues within the corresponding NUTS II region.	Orbis
f_roaa_w	Hotels return on average assets Winsorized at significance level 5%	Hotels Return on Average Assets (Net Income Winsorized at significance level 5% / (Total Assets Winsorized at significance level 5% (current year) + Total Assets Winsorized at significance level 5% (previous year))/2).	Orbis
f_mkt_share_nutsiii	Hotels market share NUTS III	Hotels ratio of each hotel's revenue to the total hotel revenues within the corresponding NUTS III region.	Orbis
f_equitycapital_total_assets	Hotels equity capital percentage to total assets	Hotel's own funds used to finance assets, excluding borrowed money for the current year (Capital / Total Assets).	Orbis
f_tot_oper_rev_turn_nutsii	Hotels total operating turnover NUTS II	Hotels total operating turnover with the corresponding NUTS II region.	Orbis
f_equitycapital_total_assets_w	Hotels equity capital percentage to total assets Winsorized at significance level 5%	Hotel's own funds are used to finance assets Winsorized at significance level 5%, excluding borrowed money of current year (Capital Winsorized at significance level 5% / Total Assets Winsorized at significance level 5%).	Orbis
f_tot_oper_rev_turn_nutsiii	Hotels total operating turnover NUTS III	Hotels total operating turnover with the corresponding NUTS III region.	Orbis
f_sector_roaa_volatility	Hotels sector return on average assets volatility	Hotels capture fluctuations in all Return on Average Assets each year (standard deviation in all Return on Average Assets current year).	Orbis
f_tot_oper_rev_turn_munic	Hotels total operating turnover municipality	Hotels total operating turnover with the corresponding municipality.	Orbis
f_roaa_volatility_w	Hotels sector return on average assets volatility Winsorized at significance level 5%	Hotels capture the fluctuations in all Return on Average Assets Winsorized at significance level 5% each year (standard deviation in all Winsorized at significance level 5% Return on Average Assets current year).	Orbis
f_mkt_share_pwr_nutsii	Hotels market share power two per NUTS II	Hotels squared market share of all hotels within each NUTS II region.	Orbis
f_mkt_share_pwr_nutsiii	Hotels market share power two per NUTS III	Hotels squared market share of all hotels within each NUTS III region.	Orbis
f_mkt_share_pwr_munic	Hotels market share power two per municipality	Hotels squared market share of all hotels within each municipality.	Orbis

Annex 03 – Data preparation

To transform the financial variables of Portugal into a panel dataset by year, the process began with importing the relevant financial variables from the Orbis database from 2010 to 2015. During this step, we specifically selected the financial variables for Portugal that were relevant to our research. Additionally, we identified the appropriate Nomenclature of Economic Activities (NACE) code for the "Accommodation" sector, which corresponds to NACE code 55. Once the data was imported, the next step involved preprocessing the datasets. This included creating a new column to record the corresponding year for each entry. The column labels were then adjusted by removing unnecessary elements, such as "EUR Year," to standardize the headers. Cells containing values such as "n.a." or "n.s." were replaced with empty values to ensure the dataset was suitable for quantitative analysis and free of errors. Finally, the affected columns were converted to numeric format to facilitate further analysis.

During this process, we identified errors in the "Opening Year" variable, where some values were recorded as "p," 38, 365, or 13. We addressed this by replacing these values with 99, indicating that they were for future analysis. Additionally, we corrected inconsistencies in the NUTS II and NUTS III regional designations. Specifically, we renamed hotels in the NUTS II regions "Grande Lisboa," "Oeste e Vale do Tejo," and "Península de Setúbal" to "Área Metropolitana de Lisboa" to align with the NUTS 2013 regional criteria. Similarly, we updated the NUTS III regions, changing "Península de Setúbal, Grande Lisboa" to "Área Metropolitana de Lisboa" and "Alto Tâmega e Barroso" to "Alto Tâmega."

Furthermore, we replaced symbolic star ratings of the hotels with actual numerical values to accurately reflect their classification.

Additionally, our focus was specifically on analyzing hotels registered as firms, i.e. "legal entities" and excluding "individuals". Only the TINs start with 5, which correspond to enterprise entities such as businesses and companies.

Following the analysis of the characteristics of Portuguese hotels, we aimed to integrate this information with financial variables obtained from the Orbis dataset. To ensure consistency between the datasets, we calculated the percentage of hotels whose Tax Identification Numbers (TINs) matched between the Turismo de Portugal dataset and the Orbis dataset, focusing on the Nomenclature of Economic Activities (NACE) codes and classified as legal entities, where TINs beginning with "5" represent corporations or business entities. The analysis revealed a 95.33% compatibility rate, indicating that most hotels in the Turismo de Portugal dataset had corresponding financial data available in the Orbis dataset. Furthermore, we verified the compatibility between the financially variable TINs and the NACE TINs, achieving a 99.78% match between the two datasets. Subsequently, we verified the compatibility between the financial variable TINs and the Portugal hotels TINs, achieving a 93.78% match between the two datasets.

This dataset is also crucial because it will allow us to integrate the financial variables. By linking the TINs of the Portuguese hotels with the NACE codes, we can merge the financial data associated with these same TINs, enabling a comprehensive analysis.

After finishing the treatment of the entire dataset of Portugal hotels with their corresponded financial variables we began primarily using Stata software, by importing the dataset and assigning descriptive labels to the respective

variables. This step included renaming the variables to ensure clarity and facilitate easy identification of their origin and purpose within the software.

Following the Winsorizing procedure, we adjusted the data types of the variables as necessary, converting categorical variables into numeric formats to enable their use in regression analysis and quantitative evaluations. Additionally, we transformed certain variables with categorical values into numeric representations to facilitate compatibility with regression models.

Table 12: 2013 NUTS II regions and NUTS III regions

Stata Variables	Dataset and Variables	Definition	Source
ht_nuts_2_oldcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos II	Hotels Nomenclature of Territorial Units for Statistical Purposes II refers to a classification system used by the European Union to divide regions for statistical analysis and reporting purposes (Alentejo, Algarve, Área Metropolitana de Lisboa, Centro, Norte, Região Autónoma da Madeira, Região Autónoma dos Açores).	Turismo de Portugal
ht_nuts_2_c_oldcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos II categories	Hotels Portugal Nomenclature of Territorial Units for Statistical Purposes II per categories numbers (1 - Alentejo; 2 - Algarve; 3 - Área Metropolitana de Lisboa; 4 - Centro; 5 - Norte; 6 - Região Autónoma da Madeira; 7 - Região Autónoma dos Açores).	Turismo de Portugal
ht_nuts_3_oldcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos III	Hotels Nomenclature of Territorial Units for Statistical Purposes III refers to a classification system used by the European Union to divide regions for statistical analysis and reporting purposes.	Turismo de Portugal
ht_nuts_3_c_oldcriteria	Hotels Nomenclatura das Unidades Territoriais para Fins Estatísticos III categories	Hotels Portugal Nomenclature of Territorial Units for Statistical Purposes II per categories numbers (1 - Alentejo Central; 2 - Alentejo Litoral; 3 - Algarve; 4 - Alto Alentejo; 5 - Alto Minho; 6 - Alto Tâmega; 7 - Área Metropolitana do Porto; 8 - Área Metropolitana Lisboa; 9 - Ave; 10 - Baixo Alentejo; 11 - Beira Baixa; 12 - Beiras e Serra da Estrela; 13 - Cávado; 14 - Douro; 15 - Lezíria do Tejo; 16 - Médio Tejo; 17 - Oeste; 18 - Região Autónoma da Madeira; 19 - Região Autónoma dos Açores; 20 - Região de Aveiro; 21 - Região de Coimbra; 22 - Região de Leiria; 23 - Tâmega e Sousa; 24 - Terras de Trás-os-Montes; 25 - Viseu Dão Lafões).	Turismo de Portugal

NUTS II regions include Alentejo, Algarve, Área Metropolitana de Lisboa, Centro, Norte, Região Autónoma da Madeira, and Região Autónoma dos Açores.

NUTS III regions further refine this classification into Alentejo Central, Alentejo Litoral, Algarve, Alto Alentejo, Alto Minho, Alto Tâmega, Área Metropolitana de Lisboa, Área Metropolitana do Porto, Ave, Baixo Alentejo, Beira Baixa, Beiras e Serra da Estrela, Cávado, Douro, Lezíria do Tejo, Médio Tejo, Oeste, Região Autónoma da Madeira, Região Autónoma dos Açores, Região de Aveiro, Região de Coimbra, Região de Leiria, Tâmega e Sousa, Terras de Trás-os-Montes, and Viseu Dão Lafões.

We categorized hotels based on their size using the number of rooms, as defined by Devesa & Peñalver (2013). Hotels with 20 or fewer rooms were classified as Micro-Hotels, those with 21 to 50 rooms as Small Hotels, those with 51 to 80 rooms as Medium Hotels, those with 81 to 120 rooms as Big Hotels, and those with more than 121 rooms as Macro-Hotels. After applying the filter for hotels opened before 2015 (registered as legal entities or as individuals), we identified 24 Micro-Hotels, 279 Small Hotels, 206 Medium Hotels, 192 Big Hotels, and 448 Macro-Hotels. In total, 509 hotels were categorized as MSMEs, representing 44% of all registered hotels, while 640 hotels were categorized as BMEs, accounting for 56% of the total.

We categorized hotels as part of a group's chain or not, i.e. a collection of hotels that operate under the same brand name and are managed by the same company. We created a binary dimension variable following the location choice of chained methodology (Yang et al., 2014). To construct this variable, we examined the Tax Identification Numbers (TINs) of Portuguese hotels. If a TIN was repeated, we assigned a value of 1, indicating that the hotel is part of a chain. Otherwise, we assigned a value of 0, signifying that it is an individual hotel.

Two distinct methodological approaches were developed to enhance the scope of the analysis. The first approach involved the creation of dummy variables based on hotel characteristics, the availability of meeting rooms (ht_meeting_rooms_d), closure periods (ht_closure_periods_d), SPA services (ht_spa_d), outdoor pools (ht_outdoor_pools_d), indoor pools (ht_indoor_pools_d), tennis courts (ht_tennis_d), gym facilities (ht_gym_d), golf courses (ht_golf_d), other facilities (ht_other_facilities_d), tourist utility services (ht_tourist_utility_d), if hotels have only 1 secondary NACE (d_secondary_code_s), 2 secondary NACES (d_secondary_code_s_2), three secondary NACES (d_secondary_code_s_3), or four secondary NACES (d_secondary_code_s_4). We applied this approach as well for the financial variables of the respective hotels, when the income is positive (f_net_income_d). This approach aimed to evaluate how individual features of Portuguese hotels may explain variations in the control variables. The second approach focused on transforming categorical variables related to hotel location and classification into numerical identifiers. These included classification status (ht_classification_status_c), the municipality (ht_municipality_c), district (ht_district_c), NUTS II and NUTS III 2013 (ht_nuts_2_c_oldcriteria, ht_nuts_3_c_oldcriteria), the number of restaurants range criteria (ht_restaurants_c), the entities responsible for regional tourism development (ht_ert_c), hotels number of secondary NACES (c_sectordiversification), and the classification of the hotel's purpose as cultural heritage (ht_classified_heritage_c). This transformation allowed for the assessment of how regional and institutional factors across different areas of Portugal may influence the Z-Score.

In addition, two supplementary variables were developed. The first variable (ht_vlg_city_oth) reflects the hotel's geographical location, specifically distinguishing whether the hotel is situated within an urban area, village or other. This variable was introduced to investigate how a hotel's location—urban versus non-urban—may influence the credit risk associated with hotels. The second variable (ht_age) and in natural logarithm (ht_ln_age) represents the age of the hotel, calculated as the difference between the time horizon years (2010, 2011, 2012, 2013, 2014, 2015) and the year in which the hotel was created. Then we created a new variable if the hotel is in continental Portugal or the islands (ht_islands_ptcont_d), and whether it falls within the micro, small and medium size group or the big and macro size group (ht_msm_bm_d).

Furthermore, we created a new binary variable, hotel service diversity. This data was obtained from the Orbis dataset. Upon examining the dataset, we observed that some hotels provide more than one service, as indicated by their having multiple Nomenclature of Economic Activities (NACE) codes, extending beyond hospitality. Our goal is to determine whether offering additional services contributes to explaining differences in credit risk of these hotels. To achieve this, we assigned a value of 1 when a secondary NACE code was present, indicating the provision of additional services. Otherwise, we assigned a value of 0, signifying that the hotel only offered hospitality services (f_d_sectordiversification_d). In addition, a complementary variable was constructed to quantify the extent of sectoral diversification based on the number of distinct NACE codes associated with each hotel, ranging from 1 to 5 (f_d_sectordiversification_qtv99). This range reflects the maximum number of NACE

codes observed in the dataset for a single hotel entity. The objective of this variable is to understand whether a higher degree of diversification in services provided by hotels is associated with increased credit risk.

Annex 04 - Z-Score

To construct the main Z-Score variable using financial data from Orbis, focusing on Tax Identification Numbers (TINs) beginning with 55, the specific financial variables required were Profit/Loss for the Period (Net Income) and Capital for the years 2010–2015, as well as Total Assets for the period 2009–2015. We had multiple datasets with varying row counts that were merged using an inner join on the Tax Identification Number in SQL Studio, retaining only the common observations. Following the merging process, the Return on Assets Average (ROAA) was calculated by dividing the Profit/Loss for the Period (Net Income) of a given year by the sum of Total Assets from that year and the previous year, with division by two applied only if Total Assets were available for both years. If Total Assets for the previous year were unavailable, the division by two was omitted. To ensure data consistency, non-applicable values (n.a.) were replaced with empty entries, and adjustments were made for instances where financial values were recorded as zero. This issue arose when either Total Assets were missing while Profit/Loss for the Period (Net Income) was available, or when Profit/Loss for the Period (Net Income) was missing while Total Assets were available. Next, to compute ROAA Volatility, the ROAA value for each year was used to calculate the standard deviation of the ROAA values across all observations for that year. For the Capital Ratio, the Capital of each hotel was divided by its Total Assets for each year. Prior to the calculations, the dataset was cleaned by substituting non-applicable values (n.a.) with empty entries to maintain consistency and accuracy. Adjustments were also made to cases where financial values were recorded as zero. If either Total Assets or Capital were missing, resulting in DIV! or 0 values, these outputs were replaced with empty entries to ensure the data remained accurate. Finally, the Z-Score variable was constructed using the previously created variables, including ROAA, ROAA standard deviation, and the Capital Ratio, with a final refinement step to address any remaining issues such as DIV! or incorrect 0 values, as these resulted from missing data in one or more variables

Annex 05 – Data - Descriptives

After finishing the treatment of the entire panel dataset of Portugal hotels with their corresponded financial variables we began primarily using Stata software for analysing, firstly we imported the variables of the datasets from hotels and financial variables of Portugal and then we assigned descriptive labels to the respective variables. This step included renaming the variables to ensure clarity and facilitate easy identification of their origin and purpose within the software. Next, we applied Winsorized at the 5% level only for the Z -Score across the datasets. This process aimed to address extreme values and reduce the influence of spurious outliers. Following the Winsorizing procedure, we adjusted the data types of the variables as necessary, converting categorical variables into numeric formats to enable their use in regression analysis and quantitative evaluations. Additionally, we transformed certain variables with categorical values into numeric representations to facilitate compatibility with regression models. Once the variables were treated and only the relevant numeric variables were selected for regression analysis, we conducted a descriptive analysis. This included examining the quartiles, mean, standard deviation, maximum, and minimum values of the variables, both without and with consideration of the time dimension over the research period (2010–2015). Finally, several new financial indicators were generated for all

datasets derived from the imported variables. Following this, a thorough analysis of missing values was conducted to evaluate the completeness and reliability of these indicators for subsequent analyses.

Annex 06 – Z-Score Correlations on Distance and Financial Variable

The correlation between f_zscore_w and location variables of hotels near beaches, such as $ht_location_beach1$ ($\rho = -0.0622$) and $beach2$ ($\rho = -0.0615$), show slight negative associations, suggesting marginally lower financial stability. Castle and church locations also display minimal correlations, with $ht_location_castle1$ ($\rho = -0.0290$) and $church2$ ($\rho = -0.0269$). Only $ht_location_church1$ shows a weak positive correlation ($\rho = 0.0518$), indicating a overall impact from hotel location.

Among the financial variables, the strongest correlation is observed with the logarithm of total assets ($f_ln_total_assets_w$, $\rho = -0.3627$), suggesting that larger hotels may face lower credit risk. Other indicators—including liquidity ($\rho = 0.0461$), efficiency ($\rho = 0.0437$), profitability ($\rho = 0.0093$), debt ($\rho = -0.0438$), and growth ($\rho = -0.0131$). These results suggest that an increase in the z-score is modestly associated with improved liquidity and efficiency, as well as a slight decrease in the levels of debt and revenue growth among hotels.

	f_zscore_w		f_zscore_w
f_zscore_w	1	f_zscore_w	1.0000
$ht_location_1$	0.0121	$f_mkt_share_conc_nutsii$	0.1007
$ht_location_2$	-0.0019	$f_mkt_share_conc_nutsiii$	0.0262
$ht_location_3$	-0.0106	$ht_room_ocup_sea_nutsii$	0.0026
$ht_location_4$	-0.0125	$ht_room_ocup_sea_nutsiii$	0.0134
$ht_location_5$	-0.0297	$ht_events_meetings$	-0.0795
$ht_location_beach1$	-0.0622	$f_ln_total_assets_w$	-0.3627
$ht_location_beach2$	-0.0615	$f_liquidity_r$	0.0461
$ht_location_beach3$	-0.0501	f_debt_r	-0.0438
$ht_location_beach4$	-0.0433	$f_efficiency_r$	0.0437
$ht_location_beach5$	-0.0362	f_profit_r	0.0093
$ht_location_castle1$	-0.029	f_growth_r	-0.0131
$ht_location_castle2$	-0.0271	$ht_room_ocup_nutsii$	-0.0581
$ht_location_castle3$	-0.0156	$ht_room_ocup_nutsiii$	-0.0278
$ht_location_castle4$	-0.0126	$ht_quality_stars$	-0.1474
$ht_location_castle5$	-0.0113	$f_mkt_share_nutsii$	-0.0171
$ht_location_church1$	0.0518	$f_mkt_share_nutsiii$	-0.0380
$ht_location_church2$	-0.0269		
$ht_location_church3$	-0.0280		
$ht_location_church4$	-0.0286		
$ht_location_church5$	-0.0289		

Figure 1: Correlation between f_zscore_w and selected Location and Financial Indicators.

Annex 07 - Descriptive - Missing Values of Hotels Characteristics and Financial Variables

For **f_debt_r**, 5,501 values are missing from a total of 7,441, indicating 73.93% missing data. For **f_efficiency_r**, 5,525 values are missing out of 7,441, resulting in 74.25% missing data. For **f_growth_r**, 6,446 values are missing out of 7,441 observations, which amounts to 86.63% missing data. For **f_liquidity_r**, 5,528 values are missing from 7,441 observations, resulting in 74.29% missing data. For **f_ln_total_assets_w**, 5,511 values are missing out of 7,441, which amounts to 74.06% missing data. For **f_mkt_share_conc_nutsii**, 1,063 values are missing from 7,441 total observations, also indicating 14.29% missing data. For **f_mkt_share_conc_nutsiii**, 1,063 values are missing out of 7,441, resulting in 14.29% missing data. For **f_mkt_share_nutsii**, 5,548 values are missing out of 7,441, also representing 74.56% missing data. For **f_mkt_share_nutsiii**, 5,548 values are missing from 7,441 observations, resulting in 74.56% missing data. For **f_profit_r**, 5,613 values are missing from a total of 7,441 observations, corresponding to 75.43% missing data. For **f_zscore_w**, 5,511 values are missing out of 7,441, representing 74.06% missing data. For all location-related variables there are no missing values, with all 7,441 observations fully reported — **ht_location_1** through **ht_location_5**, **ht_location_beach1** through **ht_location_beach5**, **ht_location_castle1** through **ht_location_castle5**, and **ht_location_church1** through **ht_location_church5** — are complete with no missing values, each covering the full set of 7,441 observations, resulting in 0.00% missing data across these variables. For **ht_room_ocup_sea_nutsii**, 1,063 values are missing from a total of 7,441, representing 14.29% missing data. For **ht_room_ocup_sea_nutsiii**, 1,219 values are missing, corresponding to 16.38% of the total observations. Regarding **ht_events_meetings** and **ht_quality_stars**, no missing values were identified, indicating full data coverage. The variables **ht_room_ocup_nutsii** and **ht_room_ocup_nutsiii** each have 1,063 missing values, corresponding to 14.29% of the dataset. Lastly, as previously mentioned, **f_debt_r** contains 5,501 missing values out of 7,441, reflecting a high rate of 73.93% incomplete data.

Table 13 - Descriptives of missing values

Variable	Missing	Total	Percent Missing
ht_events_meetings	0	7,441	0.00%
ht_quality_stars	0	7,441	0.00%
ht_room_ocup_nutsii	1,063	7,441	14.29%
ht_room_ocup_sea_nutsii	1,063	7,441	14.29%
ht_room_ocup_nutsiii	1,063	7,441	14.29%
ht_room_ocup_sea_nutsiii	1,219	7,441	16.38%
f_zscore_w	5,511	7,441	74.06%
ht_location_1	0	7,441	0.00%
ht_location_2	0	7,441	0.00%
ht_location_3	0	7,441	0.00%
ht_location_4	0	7,441	0.00%
ht_location_5	0	7,441	0.00%
ht_location_beach1	0	7,441	0.00%
ht_location_beach2	0	7,441	0.00%
ht_location_beach3	0	7,441	0.00%
ht_location_beach4	0	7,441	0.00%

ht_location_beach5	0	7,441	0.00%
ht_location_church1	0	7,441	0.00%
ht_location_church2	0	7,441	0.00%
ht_location_church3	0	7,441	0.00%
ht_location_church4	0	7,441	0.00%
ht_location_church5	0	7,441	0.00%
ht_location_castle1	0	7,441	0.00%
ht_location_castle2	0	7,441	0.00%
ht_location_castle3	0	7,441	0.00%
ht_location_castle4	0	7,441	0.00%
ht_location_castle5	0	7,441	0.00%
f_ln_total_assets_w	5,511	7,441	74.06%
f_liquidity_r	5,528	7,441	74.29%
f_debt_r	5,501	7,441	73.93%
f_efficiency_r	5,525	7,441	74.25%
f_profit_r	5,613	7,441	75.43%
f_growth_r	6,446	7,441	86.63%
f_mkt_share_nutsii	5,548	7,441	74.56%
f_mkt_share_nutsiii	5,548	7,441	74.56%
f_mkt_share_conc_nutsii	1,063	7,441	14.29%
f_mkt_share_conc_nutsiii	1,063	7,441	14.29%

Annex 08 - Descriptive Analysis of Financial Hotel Variables - with Years

The **f_debt_r**, the dataset includes 1,940 observations. In **2010**, the mean debt ratio was approximately 0.65 with a standard deviation of 0.42, with minimum and maximum values of -0.40 and 1.21. In **2011**, the mean increased to 0.86, with a standard deviation of 0.47, with minimum and maximum values of -0.37 and 2.40. The year **2012** recorded a mean of 1.02, with a higher standard deviation of 1.28 and with minimum and maximum values of -0.50 and 7.95. In **2013**, the mean declined slightly to 0.92, while variability remained similar, with a standard deviation of 0.99, with a minimum and a maximum of -3.55 and 7.50. In **2014**, the mean reached 0.99, but the standard deviation increases significantly to 6.36, with minimum and maximum values of -56.56 and 124.97. In **2015**, the mean dropped to 0.87, with a standard deviation of 2.14 and the minimum and maximum values of -12.85 and 45.48. Overall, the average debt ratio for the sample was 0.93, with a standard deviation of 4.41 and values ranging from -56.56 to 124.97. These figures suggest that, on average, hotels in Portugal are predominantly financed through debt, as the ratio values remain consistently close to 1. This indicates a higher level of financial risk, particularly from the perspective of banks when assessing credit risk to hotels. For **f_efficiency_r**, the dataset includes 1,916 observations. In 2010, the mean was -0.01 with a standard deviation of 0.18, with minimum and maximum values of -0.40 and 0.56. In 2011, the mean declined to -0.06 with a standard deviation of 0.27, and with minimum and maximum values of -1.30 and 0.37. In 2012, the mean was -0.04 with a standard deviation of 0.15, with minimum and maximum values of -0.49 and 0.31. The year 2013 recorded a mean of -0.01 with a standard deviation of 0.52, with minimum and maximum values of -0.87 and 5.58. In 2014, the mean remained at -0.01 with a standard deviation of 0.19, with minimum and maximum values of -2.30 and 1.80. In 2015, the mean increased to 0.02 with a standard deviation of 0.19, with a minimum of -3.44 and a maximum of 1.22. Over the

entire period, the overall mean was 0.00 with a standard deviation of 0.23, and minimum and maximum values of -3.44 to 5.58. These figures suggest low operational efficiency, as hotels in Portugal are generating relatively low income compared to the size of their assets. From a credit risk perspective, this reduced ability to generate returns from assets may hinder debt repayment capacity, thereby increasing the risk perceived by lenders. For **f_growth_r**, the dataset includes 995 observations. No data was recorded for **2010**. In **2011**, the mean growth ratio was -0.03 with a standard deviation of 0.22, with minimum and maximum values of -0.44 and 0.38. In **2012**, the mean rose to 0.05, with a higher standard deviation of 0.54, with minimum and maximum values of -0.99 and 2.06. The year **2013** showed a significant increase in the mean to 27.53, accompanied by a substantial increase in dispersion with a standard deviation of 190.22, with minimum and a maximum of -0.95 and 1,317.97. In **2014**, the mean stood at 0.99 with a standard deviation of 7.63 and with minimum and maximum values of -1.00 to 80.16. The year **2015** displayed an extreme increase in the mean to 1,641.98, alongside a highly dispersed distribution, with a standard deviation of 32,716.34 with minimum and maximum values of -1.21 and 796,338.10. Overall, the average growth ratio for the sample was 1,311.73, with a standard deviation of 29,229.28 and with minimum and maximum values of -1.21 and 796,338.10. For **f_liquidity_r**, the dataset includes 1,913 observations. In 2010, the mean liquidity ratio was approximately 2.26 with a standard deviation of 3.62, with a minimum and maximum values of 0.02 and 12.67. In 2011, the mean increased significantly to 10.80, followed by a much higher dispersion with a standard deviation of 53.11, with a minimum and maximum values of 0.09 and 315.79. The year 2012 recorded a mean of 4.25, with a standard deviation of 13.18, and with a minimum and maximum values of 0.02 and 96.37. In 2013, the mean increased to 7.95, while variability also increased significantly, with a standard deviation of 58.79 and with a minimum and maximum values of 0.00 and 670.54. For 2014, the mean peaked at 14.43, with a significant dispersion standard deviation of 279.19 and with a minimum and maximum values of 0.00 and 7,972.70. In 2015, the mean dropped to 4.22, with a lower but still substantial standard deviation of 24.36, and with a minimum and maximum values of 0.00 and 539.92. Overall, the average quick ratio for the sample was 8.97, with a very high standard deviation of 184.73 and values ranging from 0 to 7,972.70. For **f_ln_total_assets_w** this variable includes 1,930 observations and shows moderate variation over the years. The highest mean was observed in 2014, at 14.32, with a standard deviation of 1.71 and with minimum and maximum values of 8.44 and 16.15. In 2015, the mean remained above 14, with a minimum and maximum values of 8.27 and 16.03. Earlier years such as 2010 and 2012 reported lower averages of 12.81 and 13.43, respectively. Over the full period, the average stood at 14.21, with a standard deviation of 1.74. For **f_mkt_share_conc_nutsii**, the dataset contains 6,378 observations. In 2010, there were 1,063 observations with a mean of 0.49, a standard deviation of 0.35, and with minimum and maximum values of 0 and 1. In 2011, the mean increased to 0.73 with a standard deviation of 0.30, with minimum and maximum values of 0 and 1. The year 2012 recorded a mean of 0.32 and a standard deviation of 0.24, with minimum 0.13 and maximum 1.00. In 2013, the mean was 0.51 with a standard deviation of 0.28, with minimum and maximum values of 0.16 and 0.94. For 2014, the mean dropped to 0.10 with a standard deviation of 0.07, with minimum and maximum values of 0.03 and 0.33. In 2015, the mean was 0.07 with a standard deviation of 0.06, with minimum and maximum values of 0.02 and 0.32. Overall, the total mean was 0.37 with a standard deviation of 0.34, with a minimum 0 and maximum 1. For **f_mkt_share_conc_nutsiii**, the dataset comprises 6,378 observations. In 2010, there were 1,063 observations with a mean of 0.42, a standard deviation of 0.40, and with minimum and maximum values of 0 and 1. In 2011, the mean increased to 0.53 with a standard deviation of 0.39, with minimum and maximum values of 0 and 1. The year 2012 recorded a mean of 0.47 and a standard deviation

of 0.34, and with minimum and maximum values of 0 and 1. In 2013, the mean was 0.39 with a standard deviation of 0.29, with minimum 0 and maximum 1. For 2014, the mean declined to 0.16 and the standard deviation to 0.16, with minimum and maximum values of 0.03 and 0.84. In 2015, the mean was 0.13 and the standard deviation 0.12, with minimum and maximum values of 0.02 and 0.49. Overall, the total mean was 0.35 with a standard deviation of 0.34, with minimum 0 and maximum 1. For **f_mkt_share_nutsii**, the dataset includes 1,893 observations. In 2010, the mean market share was 0.35 with a standard deviation of 0.39, with minimum and maximum values of 0.00 and 1.00. In 2011, the mean decreased to 0.22 with a standard deviation of 0.35, and with minimum and maximum values of 0.00 to 1.00. The year 2012 recorded a mean of 0.17 with a standard deviation of 0.23, with minimum and maximum values of 0 and 1.00. In 2013, the mean further declined to 0.07 with a standard deviation of 0.18, with a minimum of 0 and a maximum of 0.97. For 2014, the mean dropped to 0.01 with a standard deviation of 0.04, with minimum and maximum values of -0.02 and 0.53. Similarly, in 2015, the mean remained at 0.01 with a standard deviation of 0.03, with minimum and maximum values of 0 and 0.53. Over the full period, the overall mean market share was 0.03 with a standard deviation of 0.11, and with minimum and maximum values of -0.02 to 1.00. For **f_mkt_share_nutsiii**, the dataset includes 1,893 observations. In 2010, the mean market share was 0.61 with a standard deviation of 0.40, with minimum and maximum values of 0 and 1. In 2011, the mean decreased to 0.47 with a standard deviation of 0.45, with minimum and maximum values of 0 and 1. In 2012, the mean further declined to 0.37 with a standard deviation of 0.36, with minimum and maximum values of 0 and 1. The year 2013 recorded a mean of 0.18 and a standard deviation of 0.25, with minimum and maximum values of 0 and 1.00. In 2014, the mean dropped to 0.03 with a standard deviation of 0.08, with minimum and maximum values of -0.29 and 0.91. Similarly, in 2015, the mean remained at 0.03 with a standard deviation of 0.07, with a minimum of 0 and a maximum of 0.70. Over the entire period, the overall mean market share was 0.07 with a standard deviation of 0.17, and values ranged from -0.29 to 1. For **f_profit_r**, the dataset includes 1,828 observations. In **2010**, the mean profitability ratio was -5.65 with a standard deviation of 25.05, with minimum and maximum values of -96.21 and 15.59. In **2011**, the mean increased to -1.18, with a higher standard deviation of 34.00 and with minimum and maximum values of -91.24 and 97.97. The year **2012** recorded a mean of -5.40, with a standard deviation of 20.11 and with minimum and maximum values of -86.24 and 21.67. In **2013**, the mean improved to -2.00, while variability remained elevated, with a standard deviation of 23.11, and with minimum and maximum values of -97.86 and 67.37. In **2014**, the mean turned positive at 1.38, with a standard deviation of 21.75, and with minimum and maximum values of -88.13 and 93.64. In **2015**, the mean increased to 5.34, with a slightly lower standard deviation of 19.58 and with minimum and maximum values of -90.74 and 72.68. Overall, the average profitability ratio for the sample was 2.60, with a standard deviation of 21.38 and with minimum and maximum values of -97.86 and 97.97. These figures suggest that, between 2010 and 2013, hotels were not generating sufficient earnings to cover their financial obligations, indicating elevated credit risk during this period. From 2014 onward, profitability improved, enhancing the hotels capacity to meet debt commitments and thereby reducing their credit risk exposure. For **f_zscore_w**, the dataset includes 1,930 observations. In 2010, the mean was 0.58 with a standard deviation of 1.41, and with minimum and maximum values of -0.31 and 6.81. In 2011, the mean slightly decreased to 0.45, with a standard deviation of 1.06, and with minimum and maximum values of -0.58 and 6.17. In 2012, the mean dropped further to 0.27, with a lower standard deviation of 0.67, and with minimum and maximum values of -0.21 and 4.56. In 2013, the mean increased to 0.40, but dispersion widened, with a standard deviation of 1.49 and minimum and maximum values of -1.28 and 14.18. The mean

declined again in 2014 to 0.28, with a standard deviation of 0.63 and a wider range between -2.29 and 12.31. In 2015, the mean was 0.36 with a standard deviation of 0.93, and with minimum and maximum values of -2.01 and 20.90. Overall, the average Z-score Winsorized at significance level 5% was 0.33, with a standard deviation of 0.87, and with minimum and maximum values of -2.29 and 20.90. These figures suggest generally low Z-scores across the hotels, with several firms falling into financially distressed zones, indicating an elevated credit risk in hotel sector. For **ht_room_occup_nutsii**, the dataset comprises 6,378 observations. In 2010, the mean occupancy rate was 0.48, with a standard deviation of 0.08, and minimum and maximum values of 0.36 and 0.61. The mean remained stable at 0.48 in 2011 and slightly decreased to 0.47 in 2012, with standard deviations of 0.09 in both years. In 2013, the mean returned to 0.48, increasing further to 0.51 in 2014 and 0.55 in 2015. Dispersion also widened over time, with the standard deviation rising to 0.12 in 2015, and the range expanding from 0.33 to 0.74. Overall, the average occupancy rate for NUTS II regions during the 2010–2015 period was 0.49, with a standard deviation of 0.10. These values indicate a progressive recovery in room occupancy in NUTS II regions, particularly from 2013 onwards. For **ht_room_occup_nutsiii**, the dataset also includes 6,378 observations. In 2010, the mean was 0.47, with a standard deviation of 0.09, and minimum and maximum values of 0.27 and 0.61. The mean slightly improved to 0.48 in 2011 but declined to 0.46 in 2012. In the following years, the occupancy rate gradually increased: 0.47 in 2013, 0.50 in 2014, and 0.54 in 2015. Standard deviation also increased over time, reaching 0.14 in 2014 and 2015, indicating greater variation across municipalities. The overall mean for the 2010–2015 period was 0.49, with a standard deviation of 0.12. These results suggest a similar trend to NUTS II, with room occupancy in NUTS III regions gradually improving, especially in the last two years of the period. For **ht_room_occup_sea_nutsii**, the dataset includes 6,378 observations. Across all years from 2010 to 2015, the average occupancy rate of seaside hotel rooms in NUTS II regions remained consistently at 0.02, with a standard deviation of 0.01. The minimum value across all years was 0.01, while the maximum gradually increased from 0.04 in 2010 to 0.06 in 2015. Despite the low average values, the stable mean and limited variation suggest that seaside hotel room occupancy in NUTS II regions showed minimal year-to-year fluctuation during the 2010–2015 period. For **ht_room_occup_sea_nutsiii**, the dataset includes 6,222 observations. Between 2010 and 2015, the average occupancy rate of seaside hotel rooms in NUTS III regions remained stable at 0.02, with a standard deviation of 0.01 throughout the period. Minimum values were consistently at 0, while maximum values varied slightly, ranging from 0.05 to 0.06. This stability indicates that seaside room occupancy in NUTS III regions showed very little variation over time, maintaining low but consistent usage rates during these years. For **ht_events_meetings**, the dataset includes 7,441 observations. The mean is 107.17 with a standard deviation of 229.00. Values range from a minimum of 0 to a maximum of 3,000. For **ht_location_1**, the dataset includes 7,441 observations. The mean is 3.44 with a standard deviation of 5.78. Values range from a minimum of 0.04 to a maximum of 44.40. For **ht_location_2**, the dataset includes 7,441 observations. The mean is 5.65 with a standard deviation of 7.83, and with minimum and maximum values of 0.08 and 53.64. For **ht_location_3**, the dataset includes 7,441 observations. The mean is 8.26 with a standard deviation of 9.48, and with minimum and maximum values of 0.24 to 58.58. For **ht_location_4**, the dataset includes 7,441 observations. The mean is 10.93 with a standard deviation of 10.90, and with minimum and maximum values of 0.55 and 67.78. For **ht_location_5**, the dataset includes 7,441 observations. The mean is 32.42 with a standard deviation of 73.37, and with minimum and maximum values of 0.79 and 308.96. For **ht_location_beach1**, the dataset includes 7,441 observations. The mean is 14.02 with a standard deviation of 14.75, and with minimum and maximum values of 0.05 and 73.83. For

ht_location_beach2, the dataset includes 7,441 observations. The mean is 17.87 with a standard deviation of 16.81, and with minimum and maximum values of 0.24 and 83.50. For **ht_location_beach3**, the dataset includes 7,441 observations. The mean is 22.37 with a standard deviation of 20.33 and with minimum and maximum values of 0.35 and 96.53. For **ht_location_beach4**, the dataset includes 7,441 observations. The mean is 34.47 with a standard deviation of 37.63, and with minimum and maximum values of 0.64 and 162.22. For **ht_location_beach5**, the dataset includes 7,441 observations. The mean is 53.18 with a standard deviation of 87.90, and with minimum and maximum values of 1.26 and 368.43. For **ht_location_castle1**, the dataset includes 7,441 observations. The mean is 87.51 with a standard deviation of 259.70, and with minimum and maximum values of 0.11 and 1,037.65. For **ht_location_castle2**, the dataset includes 7,441 observations. The mean is 103.71 with a standard deviation of 259.92. and with minimum and maximum values of 7.47 and 1,044.10. For **ht_location_castle3**, the dataset includes 7,441 observations. The mean is 144.69 with a standard deviation of 311.56, and with minimum and maximum values of 15.82 to 1,177.70. For **ht_location_castle4**, the dataset includes 7,441 observations. The mean is 167.39 with a standard deviation of 341.08, and with minimum and maximum values of from 25.98 to 1,512.19. For **ht_location_castle5**, the dataset includes 7,441 observations. The mean is 183.38 with a standard deviation of 357.41, and with minimum and maximum values of 31.78 to 1,684.60. For **ht_location_church1**, the dataset includes 7,441 observations. The mean is 7.02 with a standard deviation of 10.85, and with minimum and maximum values of 0.04 to 80.34. For **ht_location_church2**, the dataset includes 7,441 observations. The mean is 56.50 with a standard deviation of 171.59, and with minimum and maximum values of 0.08 and 680.93. For **ht_location_church3**, the dataset includes 7,441 observations. The mean is 73.05 with a standard deviation of 215.26, and with minimum and maximum values of 0.26 and 854.76. For **ht_location_church4**, the dataset includes 7,441 observations. The mean is 81.57 with a standard deviation of 230.43, and with minimum and maximum values of 0.61 and 922.07. For **ht_location_church5**, the dataset includes 7,441 observations. The mean is 87.84 with a standard deviation of 238.20, and with minimum and maximum values of 0.85 and 958.92. For **ht_quality_stars**, the dataset includes 7,441 observations. The mean is 3.30 with a standard deviation of 0.98, and with minimum and maximum values of 1 and 5.

Table 14: Descriptive Statistics – Financial Variables: Total Sample and Yearly (Table 1)

Year		f_liquidity_r	f_growth_r	f_debt_r	f_efficiency_r
2010	Obs	23	.	24	24
	mean	2.26	.	0.65	-0.01
	sd	3.62	.	0.42	0.18
	min	0.02	.	-0.40	-0.40
	max	12.67	.	1.21	0.56
2011	obs	35	15	38	36
	mean	10.80	-0.03	0.86	-0.06
	sd	53.11	0.22	0.47	0.27
	min	0.09	-0.44	-0.37	-1.30
	max	315.79	0.38	2.40	0.37
2012	obs	57.00	26	60	56
	mean	4.25	0.05	1.02	-0.04
	sd	13.18	0.54	1.28	0.15
	min	0.02	-0.99	-0.50	-0.49
	max	96.37	2.06	7.95	0.31
2013	obs	130.00	48	138	129
	mean	7.95	27.53	0.92	-0.01
	sd	58.79	190.22	0.99	0.52
	min	0.00	-0.95	-3.55	-0.87
	max	670.54	1,317.97	7.50	5.58
2014	obs	824.00	112	830	825
	mean	14.43	0.99	0.99	-0.01
	sd	279.19	7.63	6.36	0.19
	min	0.00	-1.00	-56.56	-2.30
	max	7,972.70	80.16	124.97	1.80
2015	obs	844	794	850	846
	mean	4.22	1,641.98	0.87	0.02
	sd	24.36	32,716.34	2.14	0.19
	min	0.00	-1.21	-12.85	-3.44
	max	539.92	796,338.10	45.48	1.22
Total	obs	1,913	995	1,940	1,916
	mean	8.97	1,311.73	0.93	0.00
	sd	184.73	29,229.28	4.41	0.23
	min	0	-1.21	-56.56	-3.44
	max	7,972.70	796,338.10	124.97	5.58

Note: obs denotes number of observations. sd denotes standard deviation. min denotes minimum. max denotes maximum.

Table 15: Descriptive Statistics – Financial Variables: Total Sample and Yearly (Table 2)

Year		f_ln_total_assets_w	f_mkt_share_conc_nutsiii	f_mkt_share_nutsii	f_mkt_share_conc_nutsii
2010	obs	24	1,063	23	1,063
	mean	12.81	0.42	0.35	0.49
	sd	1.92	0.40	0.39	0.35
	min	8.76	0	0.00	0
	max	15.58	1	1	1
2011	obs	37	1,063	36	1,063
	mean	13.98	0.53	0.22	0.73
	sd	1.85	0.39	0.35	0.30
	min	8.76	0	0.00	0
	max	15.58	1	1	1
2012	obs	58	1,063	54	1,063
	mean	13.43	0.47	0.17	0.32
	sd	1.87	0.34	0.23	0.24
	min	7.82	0	0	0.13
	max	15.58	1	1	1
2013	obs	131	1,063	125	1,063
	mean	13.71	0.39	0.07	0.51
	sd	1.93	0.29	0.18	0.28
	min	8.05	0	0	0.16
	max	15.87	1	0.97	0.94
2014	obs	830	1,063	811	1,063
	mean	14.32	0.16	0.01	0.10
	sd	1.71	0.16	0.04	0.07
	min	8.44	0.03	-0.02	0.03
	max	16.15	0.84	0.53	0.33
2015	obs	850	1,063	844	1,063
	mean	14.28	0.13	0.01	0.07
	sd	1.69	0.12	0.03	0.06
	min	8.27	0.02	0	0.02
	max	16.03	0.49	0.53	0.32
Total	obs	1,930	6,378	1,893	6,378
	mean	14.21	0.35	0.03	0.37
	sd	1.74	0.34	0.11	0.34
	min	7.82	0	-0.02	0
	max	16.15	1	1	1

Note: obs denotes number of observations. sd denotes standard deviation. min denotes minimum. max denotes maximum.

Table 16: Descriptive Statistics – Financial Variables: Total Sample and Yearly (Table 3)

Year		f_profit_r	f_zscore_w	f_mkt_share_nutsiii
2010	Obs	23	24	23
	Mean	-5.65	0.58	0.61
	Sd	25.05	1.41	0.40
	min	-96.21	-0.31	0
	max	15.59	6.81	1
2011	obs	36	37	36
	mean	-1.18	0.45	0.47
	sd	34.00	1.06	0.45
	min	-91.24	-0.58	0
	max	97.97	6.17	1
2012	obs	52	58	54
	mean	-5.40	0.27	0.37
	sd	20.11	0.67	0.36
	min	-86.24	-0.21	0
	max	21.67	4.56	1
2013	obs	121	131	125
	mean	-2.00	0.40	0.18
	sd	23.11	1.49	0.25
	min	-97.86	-1.28	0
	max	67.37	14.18	1.00
2014	obs	775	830	811
	mean	1.38	0.28	0.03
	sd	21.75	0.63	0.08
	min	-88.13	-2.29	-0.29
	max	93.64	12.31	0.91
2015	obs	821	850	844
	mean	5.34	0.36	0.03
	sd	19.58	0.93	0.07
	min	-90.74	-2.01	0
	max	72.68	20.90	0.70
Total	obs	1,828	1,930	1,893
	mean	2.60	0.33	0.07
	sd	21.38	0.87	0.17
	min	-97.86	-2.29	-0.29
	max	97.97	20.90	1

Note: obs denotes number of observations. sd denotes standard deviation. min denotes minimum. max denotes maximum.

Table 17: Descriptive Statistics – Financial Variables: Total Sample and Yearly (Table 4)

Year		ht_room_ocup_nutsii	ht_room_ocup_nutsiii	ht_room_ocup_sea_nutsiii	ht_room_ocup_sea_nutsii
2010	obs	1,063	1,063	1,037	1,063
	mean	0.48	0.47	0.02	0.02
	sd	0.08	0.09	0.01	0.01
	min	0.36	0.27	0	0.01
	max	0.61	0.61	0.06	0.04
2011	obs	1,063	1,063	1,037	1,063
	mean	0.48	0.48	0.02	0.02
	sd	0.09	0.10	0.01	0.01
	min	0.35	0.30	0	0.01
	max	0.63	0.63	0.05	0.05
2012	obs	1,063	1,063	1,037	1,063
	mean	0.47	0.46	0.02	0.02
	sd	0.09	0.11	0.01	0.01
	min	0.34	0.26	0	0.01
	max	0.62	0.62	0.05	0.05
2013	obs	1,063	1,063	1,037	1,063
	mean	0.48	0.47	0.02	0.02
	sd	0.10	0.12	0.01	0.01
	min	0.33	0.24	0	0.01
	max	0.65	0.65	0.05	0.05
2014	obs	1,063	1,063	1,037	1,063
	mean	0.51	0.50	0.02	0.02
	sd	0.11	0.14	0.01	0.01
	min	0.35	0.21	0	0.01
	max	0.69	0.69	0.05	0.05
2015	obs	1,063	1,063	1,037	1,063
	mean	0.55	0.54	0.02	0.02
	sd	0.12	0.14	0.01	0.01
	min	0.38	0.24	0	0.01
	max	0.74	0.74	0.06	0.06
Total	obs	6,378	6,378	6,222	6,378
	mean	0.49	0.49	0.02	0.02
	sd	0.10	0.12	0.01	0.01
	min	0.33	0.21	0	0.01
	max	0.74	0.74	0.06	0.06

Note: obs denotes number of observations. sd denotes standard deviation. min denotes minimum. max denotes maximum.

